

ME3201 Machine Design Project

Semester 5

Gearbox for fine material washer

Final Report

By

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2 Section 1 – Topic: Gearbox for yarn winding machine (210684K)

2.1 Introduction

The textile industry heavily relies on efficient and cost-effective yarn winding processes to maintain high production standards. Yarn winding involves winding yarn onto bobbins or cones, a crucial step before weaving or knitting. Currently, high-capacity motors are used to achieve the required speeds, which vary based on yarn material, thickness, and quality. Usually, this will cost a lot for the company when initializing the manufacturing plant. In this project aims to design and implement a gearbox with three forward gears and one reverse gear to improve efficiency and reduce costs associated with yarn winding machines.



2.2 Acceptance or rejection statement

The project was basically designed based on the actual machine setup that is used in yarn winding industry which we've visited during the POM project at ECO-Splinters (PVT) Ltd. Yarn winding machines typically use motors that need to operate at various speeds depending on the specific requirements of the yarn being processed. High-capacity motors are often necessary to handle peak speed demands, leading to significant initial investment costs. However, these motors frequently operate at medium speeds, resulting in under utilization and inefficiency. By integrating a gearbox, the speed can be adjusted more effectively, allowing for the use of lower-capacity motors and thus reducing initial costs as well as the maintenance cost. In the other hand, we aimed to design the reverse gear which enables the machine to stop or reverse each bobbin separately.

The project idea was accepted by all two advisors while in the problem formulation presentation. However, we didn't use it further for our project due to its complexity and the amount of factors effecting to the calculation phase are comparatively large when compared to the project that we've selected to continue. As conclusion, the project idea was good and applicable and also accepted by the advisors. But, we didn't continue with the idea because of its complexity and one of our group member had a more applicable project idea than this one.

2.3 Design Calculations and Materials Selection

From here onwards, the calculations are done for the project topic that we've selected to continue which is gearbox for fine material washer.

Selected components by 210684K:

- Couplings
- Keys and keyways
- Splines
- Lubrication

2.3.1 Coupling calculations

The design includes two different couplings to achieve the required performance from the system. At the first step, calculations were done for the coupling connecting prime mover shaft and the pulley drive 1 shaft. And then the calculations were carried out to design the coupling connecting the output shaft from in the gearbox and the screw conveyor shaft. To avoid over complicating, as the first step complete calculation was done for the first coupling and then using similar method, second coupling was analysed using excel document attached in a appendices section.

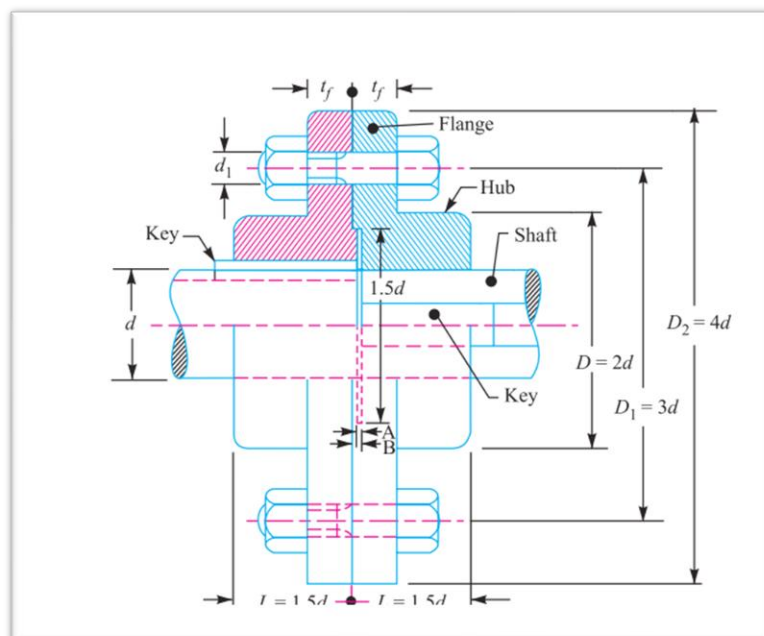
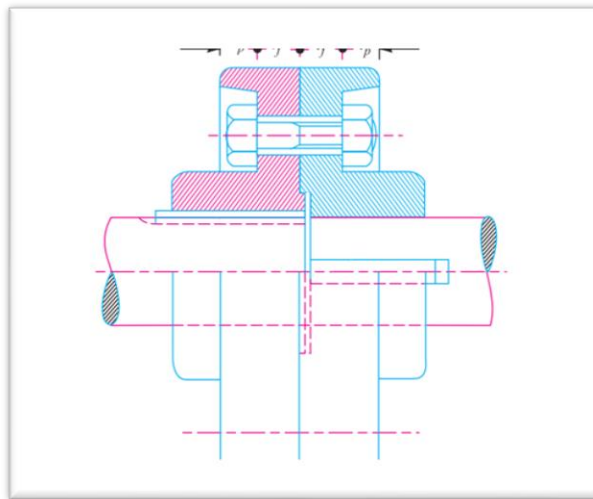
2.3.1.1 Designing prime mover and pulley driver connecting coupling

Selected type of coupling = **cast iron protected type flange coupling**

Reasons for the selection:

- **High torque capacity:** Flange couplings can transmit higher torque values which is an essential property in fine material washer machines.

- **Ease of maintenance:** Flange couplings are usually easy to assemble and disassemble, making it easy to maintain and repair in case of a breakdown.
- **Alignment and flexibility:** They provide better shaft alignment and can handle slight misalignments, which is useful in applications with varying loads, such as a friction washer with varying bottle supply.
- **Durability:** flange couplings can withstand in harsh conditions like those can be found in waste plastic recycling plant.



d: Diameter of the shaft or the inner diameter of the hub.

D: Outer diameter of the hub.

d_1 : Nominal or outside diameter of the bolt.

D_1 : Diameter of the bolt circle.

n : Number of bolts.

t_f : Thickness of the flange.

τ_s , τ_b , and τ_k : Allowable shear stress for the shaft, bolt, and key material, respectively.

τ_c : Allowable shear stress for the flange material (e.g., cast iron).

σ_{cb} and σ_{ck} : Allowable crushing stress for the bolt and key material, respectively.

Calculations:

Selected Prime mover: **Single phase motor TMY Permanent Split Capacitor 2.2kW motor**

TM series motors are low weight design and come standard with a drilled and tapped hole in the shaft which makes this series ideal for fan applications. This motor series is also ideally suited for fitting to aluminium gearboxes.

Operating parameters of the prime mover:

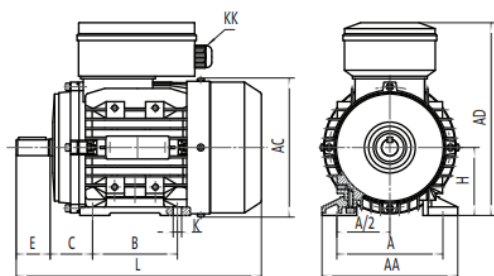
- 220V to 240V, 50Hz supply
- Continuous (S1) duty
- Ambient temperatures up to 400°C
- Installation up to 1000 MASL

Selected prime mover data:

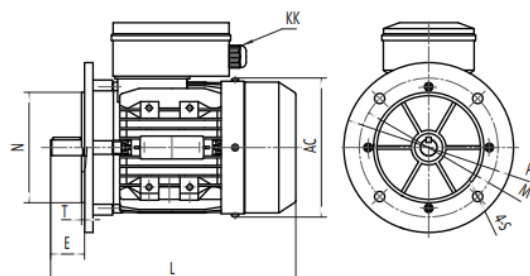
TMY DATA

Output kW	Full Load Speed [RPM]	Frame Size	Shaft Dia [mm]	Current @ 240v		Efficiency	Power Factor	Torque			Capacitor Run Capacitor [mfd/volt]	Weight Foot Mount [Kg]
				Full Load [A]	Locked Rotor [%]			Full Load [Nm]	Locked Rotor [% FLT]	Break Down [% FLT]		
0.09	2760	56A	9	0.76	380	54.0	0.94	0.3	0.65	160	4/450	2.9
0.12	2770	56B	9	0.94	410	58.0	0.92	0.4	0.65	160	6/450	3.2
0.18	2780	63A	11	1.3	380	62.0	0.95	0.6	0.60	170	10/450	4.0
0.25	2780	63B	11	1.7	400	65.0	0.95	0.9	0.60	170	12/450	4.5
0.37	2800	71A	14	2.4	400	67.0	0.95	1.3	0.60	170	16/450	5.1
0.55	2810	71B	14	3.3	430	70.0	0.98	1.9	0.55	170	24/450	7.2
0.75	2810	80A	19	4.4	430	72.0	0.98	2.5	0.35	170	25/450	9.6
1.1	2820	80B	19	6.2	430	75.0	0.98	3.7	0.33	170	35/450	11.0
1.5	2820	90S	24	8.1	460	76.0	0.98	5.1	0.30	180	45/450	14.0
2.2	2820	90L	24	12.2	470	77.0	0.98	7.5	0.30	180	60/450	16.5
0.06	1360	56A	9	0.57	420	48.0	0.92	0.4	0.75	160	4/450	3.5
0.09	1370	56B	9	0.80	360	51.0	0.92	0.6	0.75	160	6/450	3.8
0.12	1380	63A	11	0.99	340	55.0	0.92	0.8	0.65	160	10/450	4.0
0.18	1390	63B	11	1.4	370	57.0	0.92	1.2	0.65	150	10/450	4.6
0.25	1400	71A	14	1.8	420	61.0	0.94	1.7	0.50	150	14/450	5.7
0.37	1400	71B	14	2.4	360	62.0	0.94	2.5	0.50	150	16/450	6.7
0.55	1400	80A	19	3.8	380	64.0	0.95	3.8	0.35	170	20/450	8.2
0.75	1410	80B	19	4.8	400	68.0	0.95	5.1	0.33	170	25/450	9.0
1.1	1410	90S	24	6.6	440	71.0	0.98	7.5	0.33	180	40/450	14.5
1.5	1420	90L	24	8.7	440	73.0	0.98	10.1	0.33	180	45/450	16.2

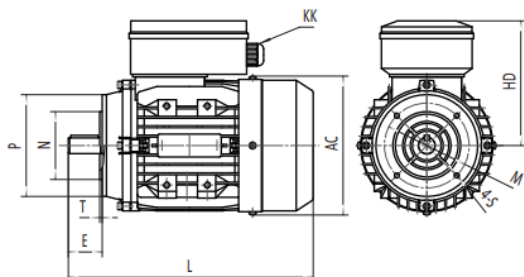
MOTOR OUTLINE DIMENSIONS



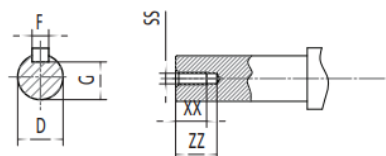
B3



B5



B14



SHAFT

TMY DATA

Frame Size	Foot Mounting				Shaft								General				
	H	A	B	C	D	E	F	G	K	SS	XX	ZZ	AA	AD	HD	AC	L
TMY 56	56	90	71	36	9	20	3	7.2	5.8	M3	9	12	110	144	88	117	196
TMY 63	63	100	80	40	11	23	4	8.5	7	M4	10	14	120	181	118	130	220
TMY 71	71	112	90	45	14	30	5	11	7	M5	12	17	132	196	125	147	241/255
TMY 80	80	125	100	50	19	40	6	15.5	10	M6	16	21	160	226	146	163	290
TMY 90S	90	140	100	56	24	50	8	20	10	M8	19	25	175	243	153	183	312
TMY 90L	90	140	125	56	24	50	8	20	10	M8	19	25	175	243	153	183	337/367
TMY 100L	100	160	140	63	28	60	8	24	12	M10	22	30	198	265	165	205	369/387

Diameter of the shaft, $d = 24 \text{ mm}$

Power transmitted by the motor shaft, $P = 2.2 \text{ kW}$

Full load speed of the motor, $N = 2820 \text{ rpm}$

Shaft material = AISI 4340 Steel

Shear strength of the shaft material = 450 MPa

Crushing strength of shaft material = 920 MPa

Shear stress for cast iron = 8 MPa

Design for hub

Determining the torque transmitted by the shaft, T

$$T = \frac{P \times 60}{2\pi N} = \frac{2.2 \times 10^3 \times 60}{2 \times \pi \times 2820} = 7.449 \text{ Nm}$$

Assume the service factor to be 1.35, therefore the maximum torque transmitted by the shaft,

$$T_{max} = 1.35 \times 7.449 = 10.056 \times 10^3 \text{ Nmm}$$

Determine the required diameter for the shaft to transmit the torque obtained above,

$$10.056 \times 10^3 = \frac{\pi}{16} \times \tau_s \times d^3 = \frac{\pi}{16} \times 450 \times d^3 = 88.357 d^3$$

$$d = 4.77 \text{ mm}$$

The actual diameter of the shaft is 24mm, so the design is safe.

Outer diameter of the hub,

$$D = 2d = 2 \times 24 = 48 \text{ mm}$$

Length of the hub,

$$L = 1.5d = 1.5 \times 24 = 36 \text{ mm}$$

Check the safety of the hub under the induced shear stress when the maximum torque transmitting while considering the hub as a hollow shaft.

$$10.056 \times 10^3 = \frac{\pi}{16} \times \tau_c \times \left[\frac{D^4 - d^4}{D} \right] = \frac{\pi}{16} \times \tau_c \times \left[\frac{48^4 - 24^4}{48} \right] = 20357.52 \times \tau_c$$

$$\tau_c = 0.4939 MPa$$

Since the induced shear stress for the hub material (i.e. cast iron) is less than the permissible value of 8 MPa, therefore the design of hub is safe.

Design for key

The keyway for a coupling has been already grooved in the motor output shaft. So, the design of the key is proceed based on the data as mentioned in the prime mover catalogue.

Width of the key, $w = 8 \text{ mm}$

Thickness of the key (D and G are provided in the catalogue),

$$t = (D - G) \times 2 = (24 - 20) \times 2 = 8 \text{ mm}$$

Length of the key, $l = \text{length of the hub, } L = 36 \text{ mm}$

Check the induced stresses in the key by considering it in shearing and crushing. Considering the key in shearing. We know that the maximum torque transmitted ($T_{\max}$),

$$10.056 \times 10^3 = l \times w \times \tau_k \times \frac{d}{2}$$

$$10.056 \times 10^3 = 36 \times 8 \times \tau_k \times \frac{24}{2}$$

$$\tau_k = 2.9097 MPa$$

Considering the key in crushing. We know that the maximum torque transmitted ($T_{\max}$),

$$10.056 \times 10^3 = l \times \frac{t}{2} \times \sigma_{ck} \times \frac{d}{2}$$

$$10.056 \times 10^3 = 36 \times \frac{8}{2} \times \sigma_{ck} \times \frac{24}{2}$$

$$\sigma_{ck} = 5.8194 MPa$$

Key material = AISI 1045 Steel, Quenched and Tempered to 450 HB

Reason:

The above type of AISI 1045 steel provides a good balance of strength, hardness, and toughness. Therefore, it is suitable for the high-stress environment of a coupling in mechanical systems.

Approximate shearing strength: 450MPa

Permissible shear stress: 225MPa

Approximate crushing strength: 310MPa

Since the induced shear and crushing stresses in the key are less than the permissible stresses, therefore the design for key is safe.

Design for flange

Flange material, similar to the hub material: cast iron

Thickness of the flange, $t_f = 0.5d$

$$t_f = 0.5d = 0.5 \times 24 = 12mm$$

check the induced shearing stress in the flange by considering the flange at the junction of the hub in shear. We know that the maximum torque transmitted (T_{max}),

$$10.056 \times 10^3 = \frac{\pi D^2}{2} \times \tau_c \times t_f = \frac{\pi (48)^2}{2} \times \tau_c \times 12 = 43429.37 \times \tau_c$$

$$\tau_c = 0.2315 MPa$$

Since the induced shear stress in the flange is less than 8 MPa, therefore the design of flange is safe.

Design for bolts

Nominal diameter of bolts = d_1

Bolt material is similar to the key material. So, the permissible shear stress = 225MPa

Selecting the number of bolts,

shaft diameter (mm)	up to 40mm	40mm to 100mm	100mm to 180mm
No. of bolts	3	4	6

No. of bolts, $n = 3$

Pitch circle diameter of bolts,

$$D_1 = 3d = 3 \times 24 = 72mm$$

The bolts are subjected to shear stress due to the torque transmitted. We know that the maximum torque transmitted (T_{max}),

$$10.056 \times 10^3 = \frac{\pi d_1^2}{4} \times \tau_b \times n \times \frac{D_1}{2} = 38170.35 \times d_1^2$$

$$d_1^2 = 0.5269$$

$$d_1 = 0.7258mm$$

This value is the diameter for permissible stress value, therefore probably we need to use a bolt with higher than this.

The nearest standard size of bolts is **M1.6**

Outer diameter of the flange,

$$D_2 = 4d = 4 \times 24 = 96mm$$

Thickness of the protective circumferential flange,

$$t_p = 0.25d = 0.25 \times 24 = 6mm$$

2.3.1.2 Repeating sample calculations for coupling 2

	Units	Coupling 1	Coupling 2
General parameters			
transmitting power	W	2200	2200
Speed in rpm	rpm	2820	15.665
transmitting torque	Nm	7.449805847	1341.107723
service factor	—	1.35	1.35
maximum transmitting torque	Nmm	10057.23789	1810495.427
permissible shear stress of shaft	MPa	287.1	287.1
permissible crushing stress of shaft	MPa	1475	1475
permissible shear stress for hub material	MPa	8	8
design for hub			
diameter of the shaft (nominal)	mm	5.62952666	31.7866517
diameter of the shaft (used)	mm	24	65
outer diameter of the hub	mm	48	130
Length of the hub	mm	36	97.5
induced shear stress at Tmax	MPa	0.494030594	4.476784702
safe unsafe condition		safe	safe
design for key			
width of the key	mm	8	20
thickness of the key	mm	8	12
length of the key	mm	36	97.5
allowable maximum shear stress	MPa	225	225
allowable maximum crushing stress	MPa	310	310
induced maximum shear stress	MPa	2.910080409	28.56797517
induced maximum crushing stress	MPa	5.820160818	95.22658391
safe unsafe condition		safe	safe
design for flange			
thickness of the flange	mm	12	32.5
induced maximum shear stress in the flange	MPa	0.231576841	2.098492829
safe unsafe condition		safe	safe
design for bolts			
number of bolts	—	3	4
pitch circle diameter of bolts	mm	72	195
nominal diameter of bolts	mm	0.725924216	5.125428894
taken bolt type	—	M1.6	M6
outer diameter of the flange	mm	96	260
protective circumferential flange thickness	mm	6	16.25
*bolt material is same as key material			
Material selection			
shaft material		AISI 4340 Steel	AISI 4340 Steel
Hub material		cast iron	cast iron
key material		cast iron	cast iron

2.3.2 Keys and keyways calculations

2.3.2.1 Sample calculations

Key is a piece of mild steel inserted between the shaft and hub or gear wheel to connect those together to prevent relative motion between them. It is always inserted parallel to the axis of the shaft. Keys are used as temporary fastenings and are subjected to considerable crushing and shearing stresses.

There are different calculations carried out for keys used in different parts of the gearbox as listed below,

- Pulley drive 1 and shaft
- Pulley drive 2 and gear box input shaft
- Gearbox input shaft and pinion gear
- Layshaft and gears mounted to lay shaft
- Idler gear shaft and idler gear wheel

First a sample calculation were done in order to understand the method of performing the calculation to obtain the dimensions of the key and then, repeating similar steps for all the other keys using excel sheet attached in the appendices section at the end of this report.

Calculations for lay shaft and gear wheels mounted to lay shaft

Key material: AISI 1018 steel

- Permissible shearing stress = 220MPa
- Permissible crushing stress = 370MPa
- Shaft diameter = 55mm
- Torque transmits in the shaft = 670.553 Nm

Width and thickness of the key can be obtained according to the shaft diameter as listed in the table below,

Shaft diameter (mm) upto and including	Key cross-section		Shaft diameter (mm) upto and including	Key cross-section	
	Width (mm)	Thickness (mm)		Width (mm)	Thickness (mm)
6	2	2	85	25	14
8	3	3	95	28	16
10	4	4	110	32	18
12	5	5	130	36	20
17	6	6	150	40	22
22	8	7	170	45	25
30	10	8	200	50	28
38	12	8	230	56	32
44	14	9	260	63	32
50	16	10	290	70	36
58	18	11	330	80	40
65	20	12	380	90	45
75	22	14	440	100	50

For 55mm shaft width of the key, $w = 18$ mm, thickness of the key, $t = 11$ mm

Let, length of the key = l

Considering the shearing of the key,

$$T = l \times w \times \tau \times \frac{d}{2}$$

$$670.553 \times 10^3 = l \times 18 \times 220 \times \frac{55}{2}$$

$$l = 6.1575 \text{ mm}$$

Considering the crushing of the key,

$$T = l \times \frac{t}{2} \times \sigma_c \times \frac{d}{2}$$

$$670.553 \times 10^3 = l \times \frac{11}{2} \times 370 \times \frac{55}{2}$$

$$l = 11.9821 \text{ mm}$$

Taking larger of the two values, we have length of key, say $l = 14$ mm

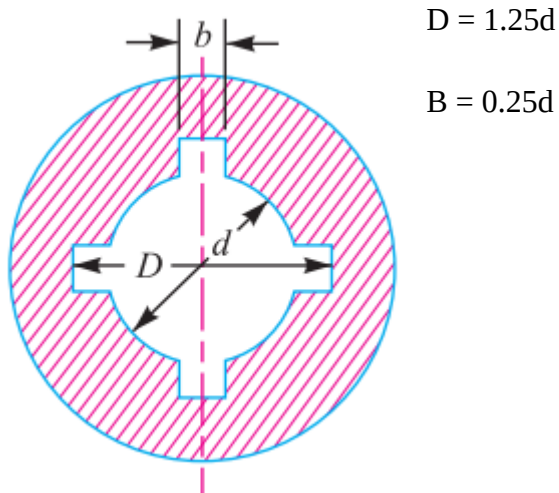
Similarly, the calculations were carried out for the other keys listed above

2.3.2.2 Calculations for other keys

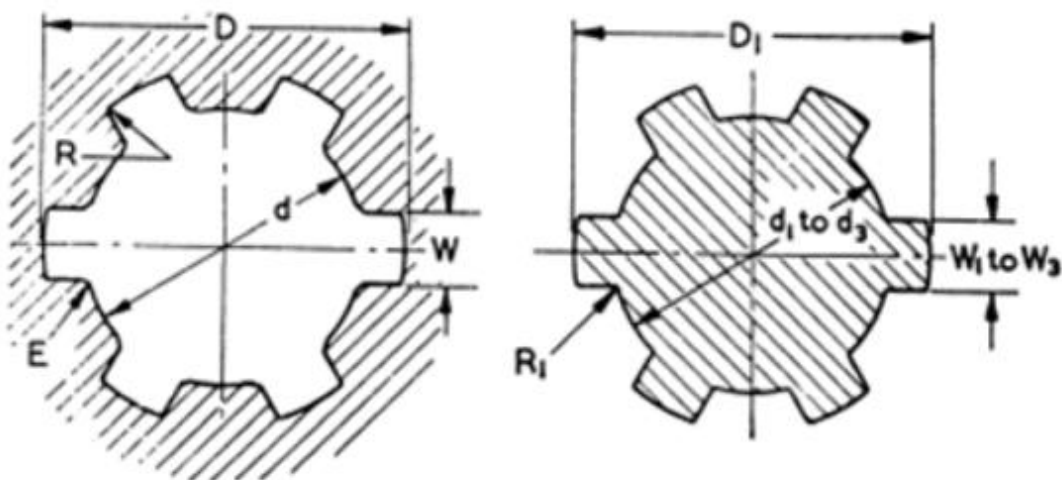
	Name	shaft diameter (mm)	w (mm)	t (mm)	Torque transmitting in the shaft (Nmm)	Permissible shear stress (MPa)	Permissible crushing stress (MPa)	Length considering shear (mm)	Length considering crushing (mm)	Length (mm)
1	Pulley drive 1 and shaft	24	10	8	10056	220	370	0.380909091	0.566216216	3
2	Pulley drive 2 and gear box input shaft	30	10	8	134102	220	370	4.06369697	6.040630631	8
3	Gearbox input shaft and pinion gear	30	10	8	134102	220	370	4.06369697	6.040630631	8
4	Layshaft and gears mounted to lay shaft	55	18	11	670553	220	370	6.157511478	11.9821845	14
5	Idler gear shaft and idler gear wheel	55	18	11	670553	220	370	6.157511478	11.9821845	14

2.3.3 Splines calculations

Sometimes, keys are made integral with the shaft which fits in the keyways broached in the hub. Such shafts are known as splined shafts.



Parallel side or straight sided splines are used in the gear box as it's the most common, easy to manufacture and easy to install spline type in the industry.



Design data

$$d = 65 \text{ mm}$$

$$D = 81.25 \text{ mm} = 82 \text{ mm}$$

$$b = 16.25 \text{ mm}$$

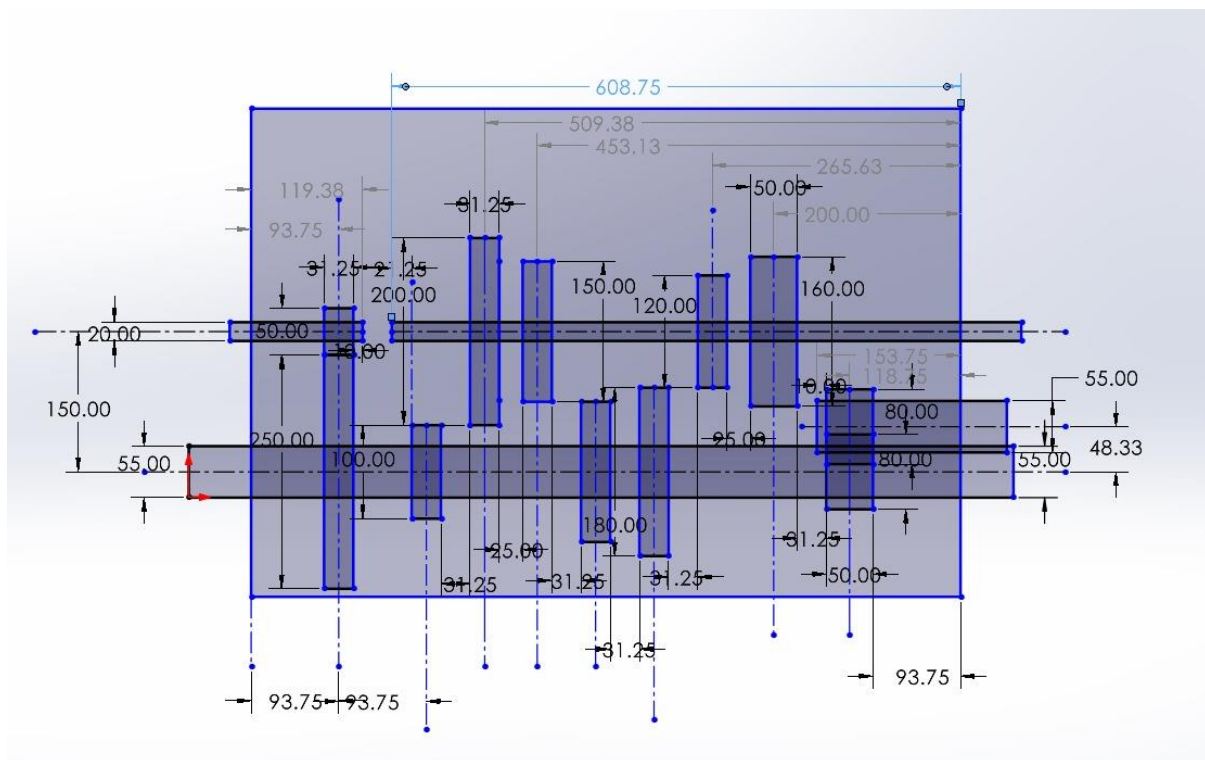
2.3.4 Lubrication calculations

Proper lubrication is crucial for a gearbox to ensure its operation is reliable and efficient. It reduces friction and wear, dissipates heat, and maintains the integrity of internal components. The importance of adopting effective lubrication methods for gearbox health and dependability cannot be overemphasized, since it has a direct impact on performance and life span. This overview looks at the most important reasons why effective lubrication is important in mechanical systems.

There are three types of lubrications

1. Grease lubrication
2. Splash (Oil bath) lubrication
3. Forced lubrication.

Gearbox configuration



Gear	Pitch line velocity
1	0.410139
2	0.410139
3	0.1640435

4	0.1640435
5	0.246065
6	0.246065
7	0.29527
8	0.29527
9	0.1312348
10	0.1312348
11	0.1312348

According to the tangential speeds we've to select the lubrication type

No.	Lubrication	Range of tangential speed v (m/s)					
		0	5	10	15	20	25
1	Grease lubrication						
2	Splash lubrication						
3	Forced oil circulation lubrication						

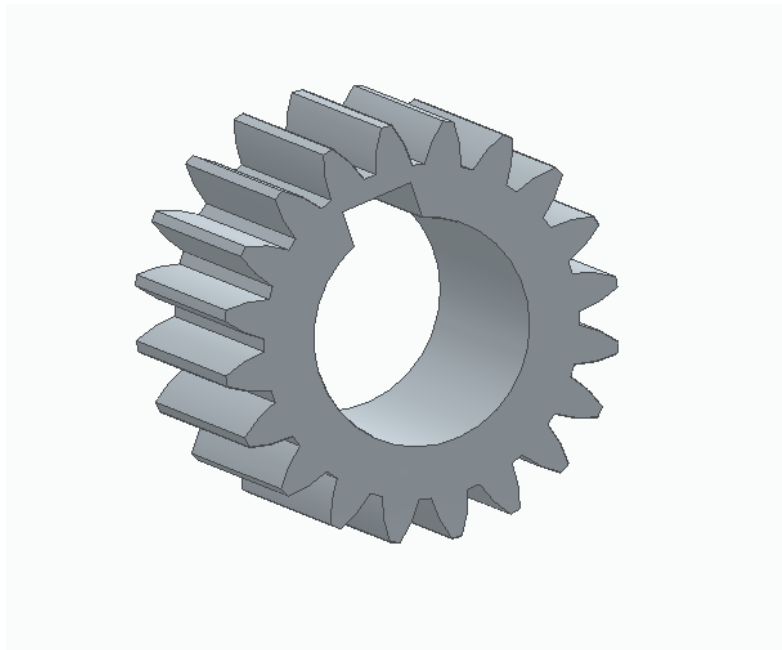
Selected lubrication type: **Grease lubrication**

2.4 Final design

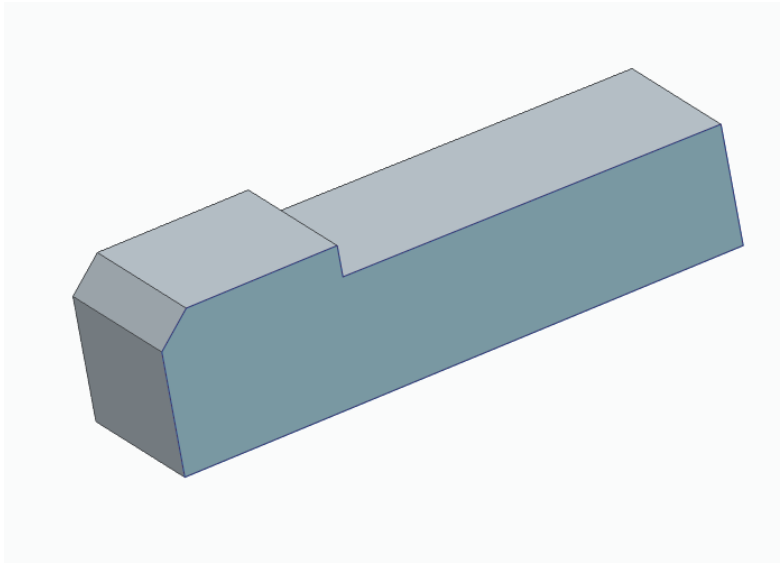
Input shaft, Input pinion gear, key and assembly



Input shaft



Input pinion gear



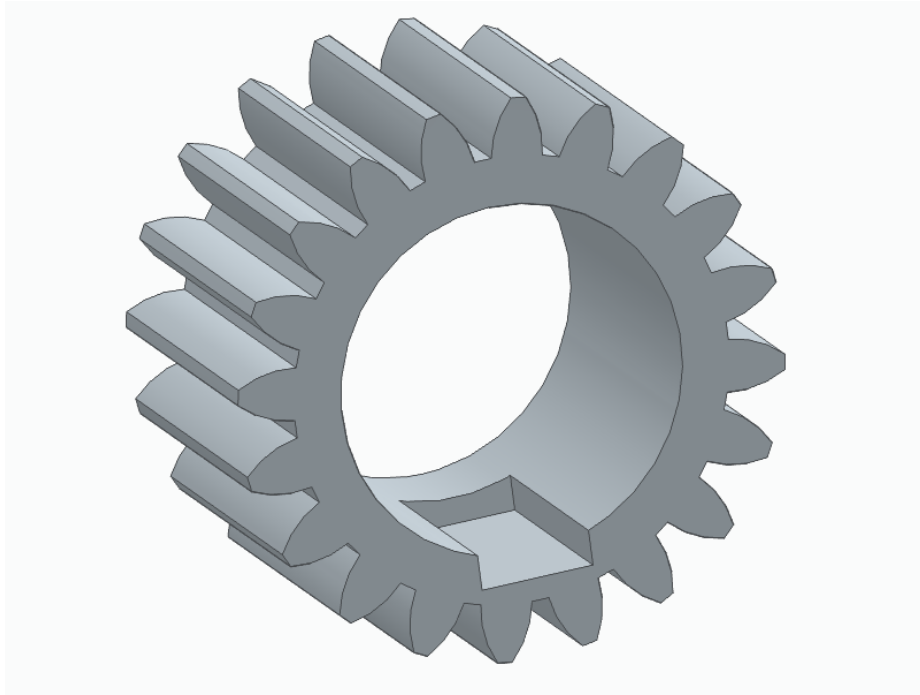
Input gear key



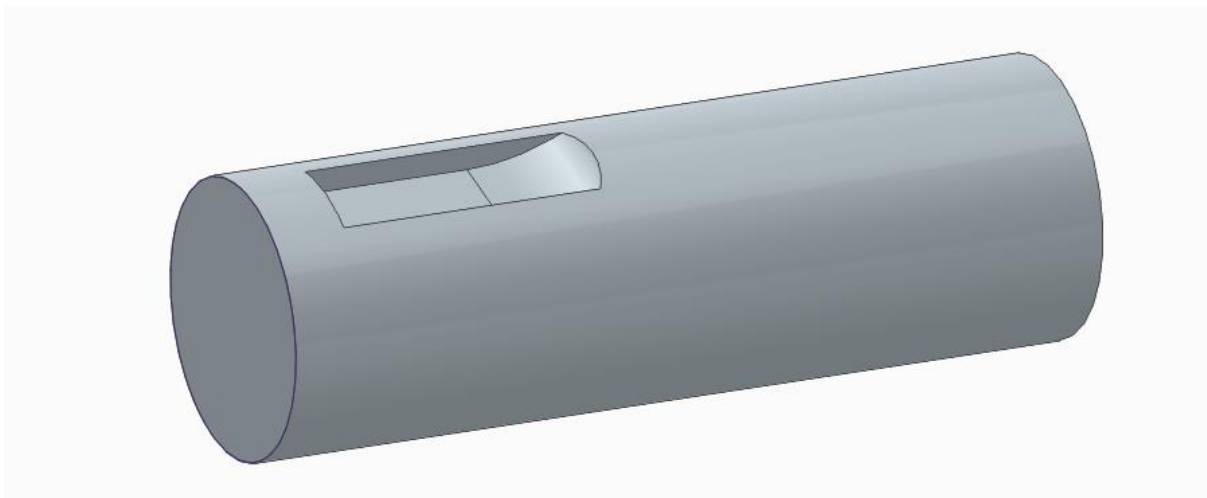
Input assembly

Input shaft is the shaft that transmit the power from the prime mover side to the gear box. A single key was used to avoid axial movements of the gear and from the other side the gear is fixed by using a step in the input shaft. This design ensure the shaft is rigidly connected to the gear and no any axial movements while the power transmission.

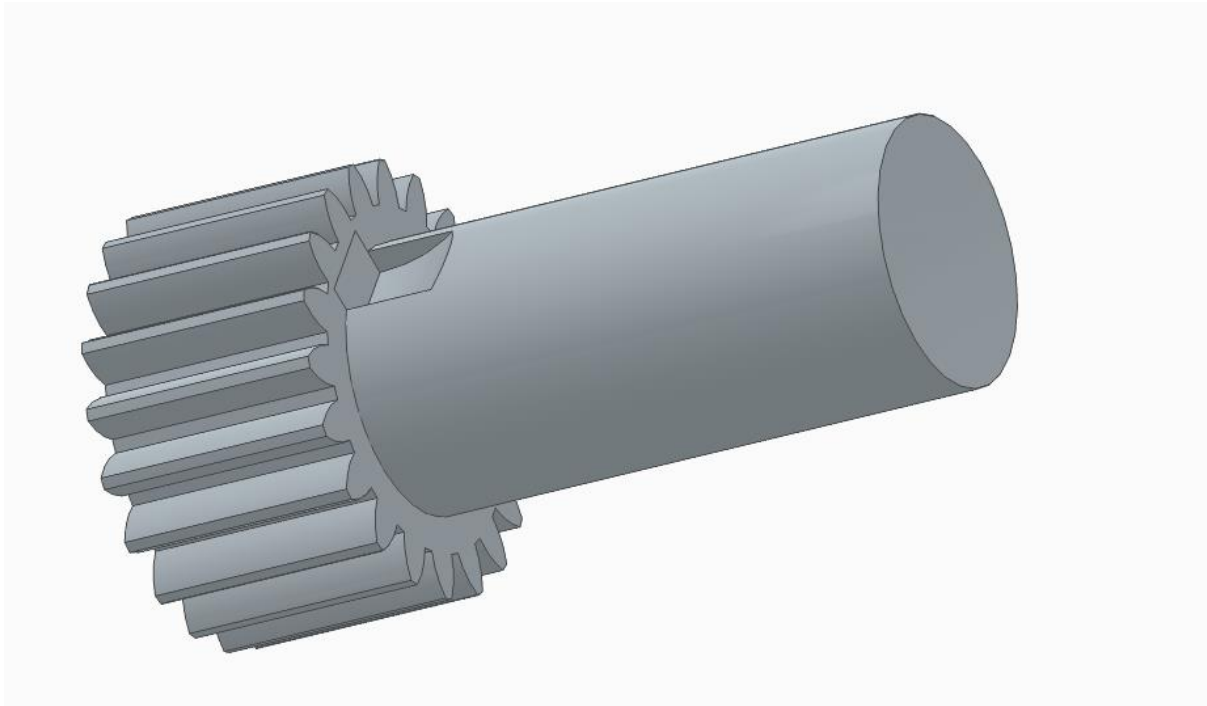
Idler gear, shaft, key and assembly



Idler gear



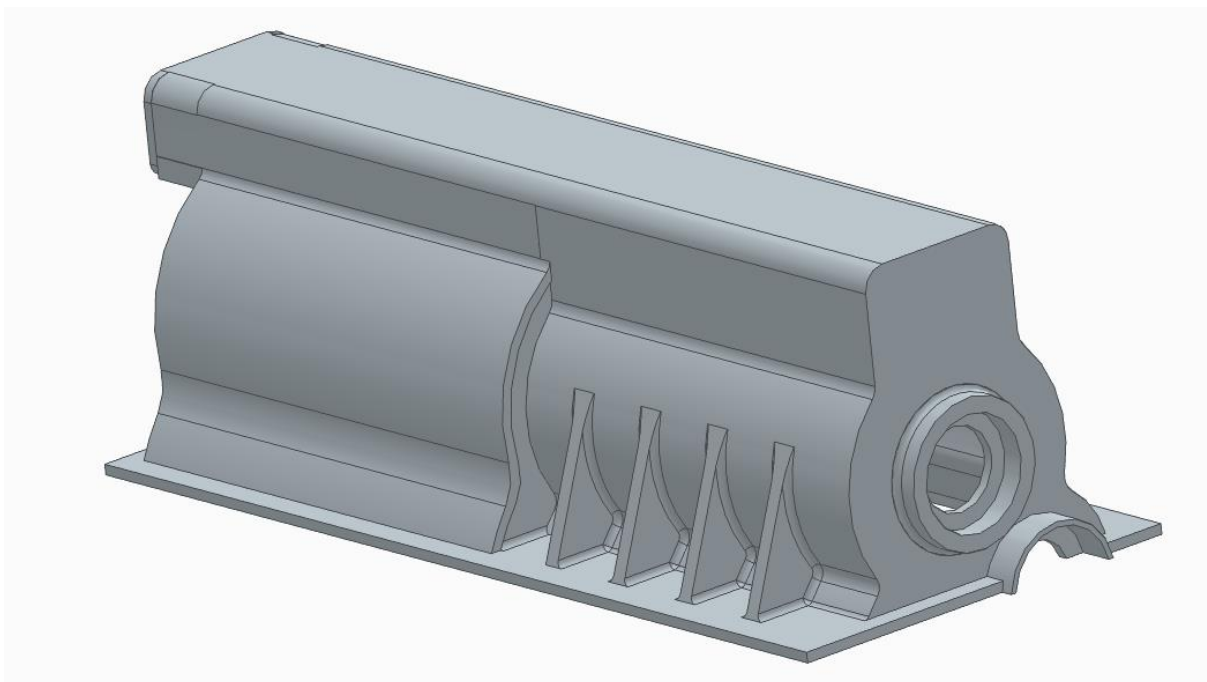
Idler shaft

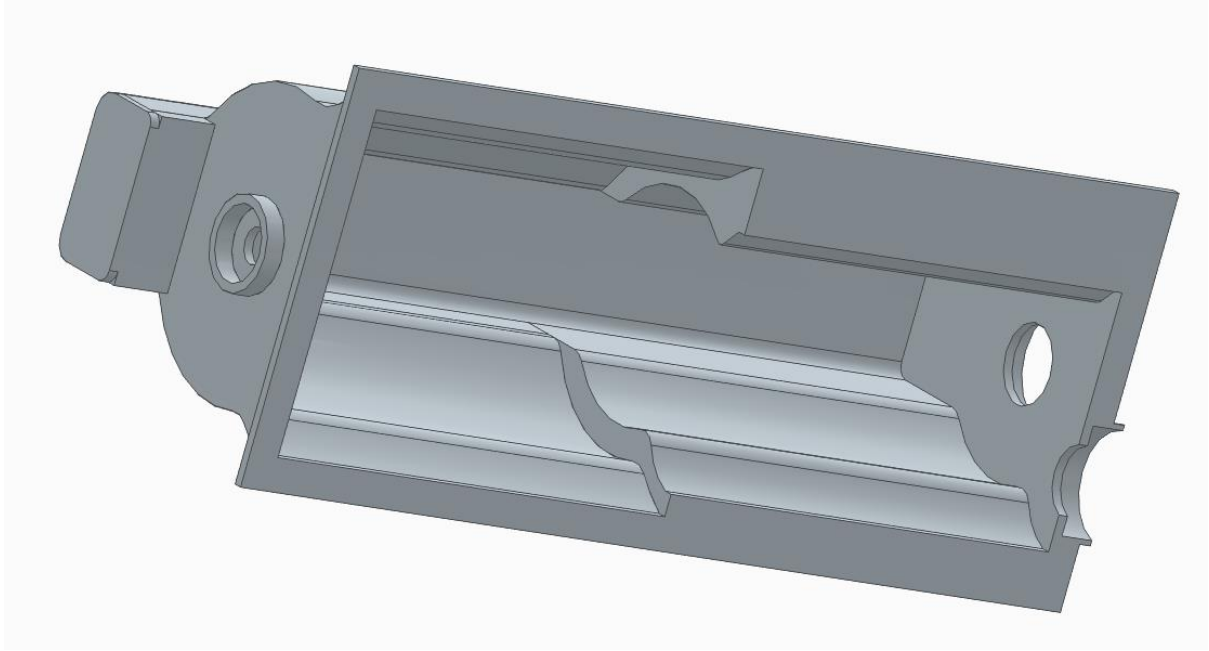


Idler gear shaft key assembly

The idler gear is the gear that is used to activate in reverse gear. The shaft is consist with a key in order to fix the gear wheel with the shaft.

Housing top part





The housing is designed to cover the upper part of the gear box. The shape of the housing was designed to minimize the material usage while supplying the required room for lubrication. The top most part of the housing is used to insert the shifting mechanism. The distance travelled during the shifting is also considered when designing the housing top part.

2.5 Future works

Currently the gears are not topology optimized. As a future step, we can perform the topology optimization for the gear wheels and reduce the material usage for the gears.

Change the housing material into more cost effective material.

2.6 References

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- [2] Gearboxman, "Introduction to Gearbox Design and Implementation," [Online]. Available: <https://www.gearboxman.com/gearbox-design>. [Accessed: 21-Jul-2024].
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- [5] ResearchGate, "Improving Energy Efficiency in Textile Manufacturing," [Online]. Available: https://www.researchgate.net/publication/319380027_Improving_energy_efficiency_in_textile_manufacturing. [Accessed: 21-Jul-2024].

3 Section 2 – Topic: Gear box for small scale waste compacting machine (210448R)

3.1 Introduction

Currently small-scale waste compacting machines are used to compact (bale) the waste materials. We are trying to implement gearbox for this compacting operation of small-scale vertical trash compactor. Based on the type of waste material, different torques should be provided to compress the waste to specific level.

Usually for this machines, different types of waste materials are coming. So, we need to apply different types of forces to compress this waste depend on various factors like weight of the waste lot and volume. So, using variable speed trash compacting machine we can control torque applied to the compaction mechanism precisely. This enables the machine to adapt to different types and densities of waste materials more effectively, ensuring optimal compaction without overload or inefficiency



3.2 Acceptance or rejection statement

The project was accepted by the advisors as it's a practical problem that can be solved using a gearbox. But because there are some limitations in using the reverse gear for the setup therefore the project was not continued with this project idea.

3.3 Design Calculations and Materials Selection

From here onwards, the calculations are done for the project topic that we've selected to continue which is gearbox for fine material washer.

Selected components by 210448R:

- Gear calculations
- Bearings

3.3.1 gear calculations

Calculation of gear reduction ratios

Motor Specifications

Parameter	Value	Unit
Power	2200	W
Speed	2820	rpm
Torque	7.5	Nm

Our required Overall reduction

Gear	Required Output speed (rpm)	Required gear reduction
1 st gear	15.75	179.04
2 nd gear	31.49	89.55
3 rd gear	47.24	59.69
Reverse gear	15.75	179.04

Reduction at Pulley drive 1

Output speed (rpm)	626.667
Reduction	4.5 : 1

Reduction at Pulley Drive 2

Output speed (rpm)	156.66
Reduction	4. : 1

Reduction between Input shaft and lay shaft = 5:1

Required Gear reduction at lay shaft and output shaft

Gear	Required Output speed (rpm)	Required gear reduction
1 st gear	15.75	1.9892:1
2 nd gear	31.49	1::1.0051
3 rd gear	47.24	1::1.5078
Reverse gear	15.75	1.9892:1

Assumptions

- We used 20 full depth involute gears
- Power losses were neglected to add additional safe factor. So we used motor power (2200W) for all the calculations
- Gears are ordinary cut
- Face width of all the gears assumed to be 12.5m (m = Module)

Module selection and validation

First we assumed different standard modules (1,2..) and checked strength of the gears using applied tangential load, static tooth load and wear tooth load. And also, we checked the dimension of the gears and gearbox. According to that we selected **2.5 mm** module for 1st gear, 2nd gear and 3rd gear and **4mm** module for reverse gear

S. No.	Particulars	14½° composite or full depth involute system	20° full depth involute system	20° stub involute system
1.	Addendum	1m	1m	0.8 m
2.	Dedendum	1.25 m	1.25 m	1 m
3.	Working depth	2 m	2 m	1.60 m
4.	Minimum total depth	2.25 m	2.25 m	1.80 m
5.	Tooth thickness	1.5708 m	1.5708 m	1.5708 m
6.	Minimum clearance	0.25 m	0.25 m	0.2 m
7.	Fillet radius at root	0.4 m	0.4 m	0.4 m

Gears (Pinion and wheel)	Module (mm)
Input and lay	2.5
1 st gear	2.5
2 nd gear	2.5

3 rd gear	2.5
Reverse gear (on lay shaft, idle and output)	4

Addendum	2.5 mm
Deddendum	3.125 mm
W.depth	5 mm
Minimum total depth	5.625 mm
Tooth thickness	3.927 mm
Minimum clearance	0.625 mm
Fillet radius at root	1 mm

Addendum	4 mm
Deddendum	5 mm
W.depth	8 mm
Minimum total depth	9 mm
Tooth thickness	6.2832 mm
Minimum clearance	1 mm
Fillet radius at root	1.6 mm

For module 2.5 gears

For module 4 gears

Minimum gear teeth calculation

The number of teeth in pinion in order to avoid interference, is obtained using following equation

$$T = \frac{2A_w}{\left\{ [G^2 + \sin^2 \psi (1 + 2G)]^{\frac{1}{2}} - G \right\}}$$

$$G = \text{Gear ratio} = \frac{\text{rpm of the pinion}}{\text{rpm of the gear}}$$

ψ = Pressure angle = 20 degrees

$A_w = 1$ (For 20 full depth involute system)

	Gear ratio
--	------------

Input: Lay	5
1st	1.9892
2nd	1.0051
3rd	1.5078
Reverse: idle	1
Idle: output	1.9892

The minimum teeth of pinion of input shaft $T_P = \frac{2 \times 1}{\left\{ [5^2 + \sin^2 20(1 + 2 \times 5)]^{\frac{1}{2}} - 5 \right\}} = 15.74045824$

The minimum teeth of wheel of lay shaft $T_G = G \times T_P = 5 \times 16 = 80$

T_P	minimum (Minimum teeth count of pinion)	T_P min (real)	Our selected Gear ratio between lay shaft and output shaft	T_G min
Input: Lay	15.74045824	16	5	80
1st	14.14844071	15	2	30
2nd	12.33817122	13	1	13
3rd	13.47165262	14	1.5	21
Reverse	12.32311873	13	1	13
Idle gear	14.14844071	15	2	30

Determining centre distance

We calculated centre distance between input shaft and lay shaft using diameters of input pinion gear and gear of lay shaft. First, we assumed reasonable number of teeth for the input pinion gear and layshaft gear and calculated centre distance. Actually, we did this for number of times and checked the actual gear ratios that we can obtained with this centre distance and our selected modules

	Selected Nu of teeth	Module (mm)	P.C.D (mm)
Pinion at input shaft	20	2.5	50
Gear at lay shaft	100	2.5	250

$$\text{Centre distance} = \frac{50+250}{2} = 150 \text{ mm}$$

Then according to centre distance and our required gear ratios we selected nu of teeth for other gears as well.

	Nu of teeth at pinion	Module	Required gear ratio between lay shaft and output	Actual gear ratio between lay shaft and output shaft	Nu of teeth at gear	P.C.D (pinion)	P.C.D (gear)	Centre distance (mm)
1st	40	2.5	1.9892	2	80	100	200	150
2nd	60	2.5	1.0051	1	60	150	150	150
3rd	48	2.5	1.5078	1.5	72	120	180	150
Reverse: idle	20	4	1	1	20	80	80	48.33
Idle: Gear	20	4	1.9892	2	40	80	160	101.67

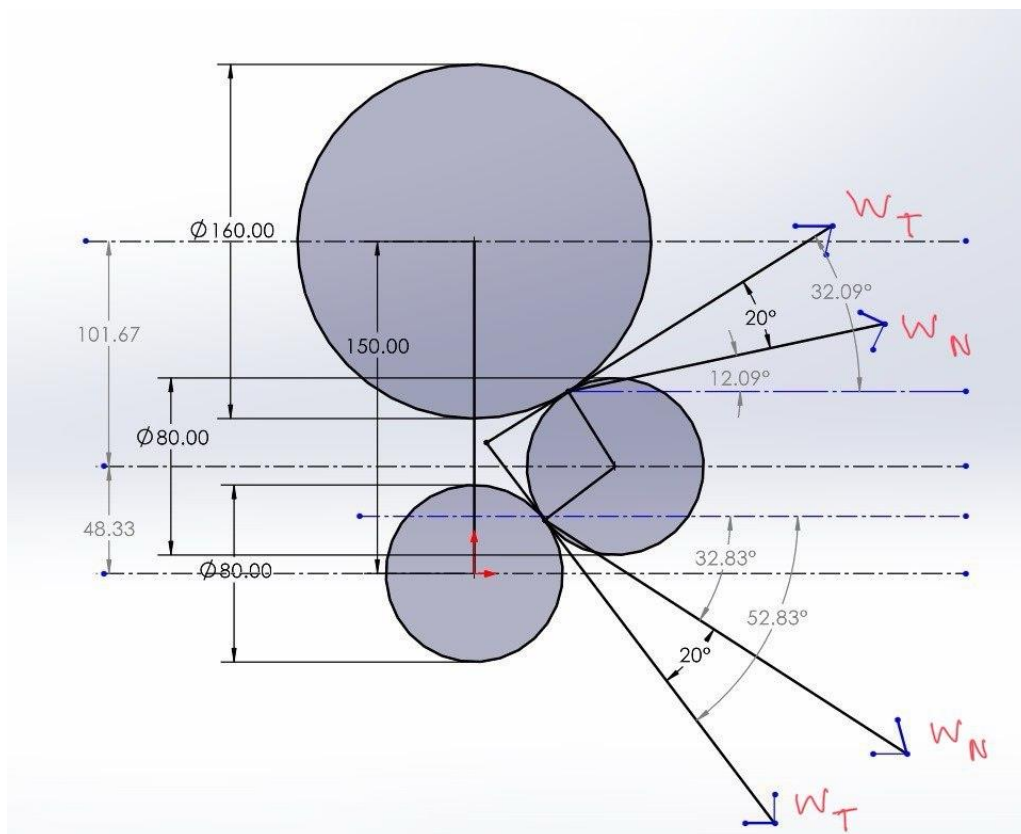
In **Reverse: idle:**

Pinion	Gear
On lay shaft	On idle shaft

In **Idle**: gear:

Pinion	Gear
On lay shaft	On idle shaft

Side view of reverse gear set



	Required output speed	Actual output speed
1 st gear	15.75	15.665

2 nd gear	31.49	31.33
3 rd gear	47.24	46.995
Reverse gear	15.75	15.665

Material selection

We used AISI 4337 steel (UNS G43370) and High carbon steel as gear materials

AISI 4337 steel (UNS G43370)

AISI 4337 Steel (UNS G43370)

Categories: [Metal](#); [Ferrous Metal](#); [Alloy Steel](#); [AISI 4000 Series Steel](#); [Low Alloy Steel](#)

Key Words: UNS G43370

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like in

 [Printer friendly version](#)  [Download as PDF](#)  [Download to Excel \(requires Excel and Windows\)](#)

 [Export data to your CAD/FEA program](#)

Physical Properties	Metric
Density	7.75 g/cc

Mechanical Properties	Metric
Hardness, Brinell	435
Tensile Strength, Ultimate	1448 MPa
Tensile Strength, Yield	965 MPa @Strain 0.2 %
Elongation at Break	14 %
Modulus of Elasticity	201 - 209 GPa
Bulk Modulus	160 - 170 GPa
Poissons Ratio	0.27 - 0.30
Shear Modulus	81.0 - 82.0 GPa

Overview of materials for High Carbon Steel

Categories: [Metal](#); [Ferrous Metal](#); [Carbon Steel](#); [High Carbon Steel](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category report the average value, and number of data points used to calculate the average. The values are based on processing methods.

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like to be listed.

 [Printer friendly version](#)  [Download as PDF](#)  [Download to Excel \(requires Excel and Windows\)](#)

 [Export data to your CAD/FEA program](#)

Physical Properties

	Metric
Density	0.451 - 8.26 g/cc
Particle Size	6.70 - 12.0 µm

Mechanical Properties

	Metric
Hardness, Brinell	163 - 600
Hardness, Knoop	195 - 769
Hardness, Rockwell B	43.0 - 100
Hardness, Rockwell C	10.0 - 70.0
Hardness, Vickers	182 - 748
Tensile Strength, Ultimate	161 - 3200 MPa
Tensile Strength, Yield	275 - 3340 MPa
Elongation at Break	0.500 - 30.0 %
Reduction of Area	13.4 - 73.0 %
Modulus of Elasticity	13.8 - 235 GPa
Flexural Yield Strength	159 - 5130 MPa
Compressive Yield Strength	1320 - 3100 MPa
Bulk Modulus	160 GPa
Poissons Ratio	0.280 - 0.313

High carbon steel

Gears and gear materials

	Input gear	Layshaft gear	1st gear		2nd gear	
	1 (pinion)	2(gear)	3 (Pinion)	4(gear)	5 (gear)	6(Pinion)
Module	2.5	2.5	2.5	2.5	2.5	2.5
Material	High carbon steel	AISI 4337 steel	High carbon steel	High carbon steel	AISI 4337 steel	AISI 4337 steel

	3rd gear		Reverse gear			
	7 (gear)	8(pinion)	9(reverse)	10(idle)	10 idle()	11 (gear)
Module	2.5	2.5	4	4	4	4
Material	AISI 4337 steel	AISI 4337 steel	High carbon steel	High carbon steel	High c steel	High c steel

Module validation using Lewis's equation

Applied tangential load on tooth

$$W_T = \frac{P}{V} \times C_s$$

- P =Power
- V= Pitch line velocity
- C_s = Service factor

We took our type of load as steady and it works continuous 24 hours per day and then

$$\text{Service factor } C_s = 1.25$$

Type of load	Type of service		
	Intermittent or 3 hours per day	8-10 hours per day	Continuous 24 hours per day
Steady	0.8	1.00	1.25
Light shock	1.00	1.25	1.54
Medium shock	1.25	1.54	1.80
Heavy shock	1.54	1.80	2.00

Allowable tangential load on tooth using Lewis equation

$$W_T = \sigma_w \times b \times \pi \times m \times y$$

- σ_w = Working stress
- b = Face width =12.5*m
- m =Module
- y = Lewis tooth form factor
- C_v = Velocity factor
- σ_0 = Allowable static stress
- $\sigma_w = \sigma_0 \times C_v$
- $y = \text{Lewis form factor} = 0.154 - \frac{0.912}{T}$ For 20 Full depth involute systems
- $C_v = \text{Velocity factor} = \frac{3}{(3+V)}$ for ordinary cut gears

For High carbon steel

- Max UTS (MPa) = 3200
- Allowable static stress = $\frac{\text{Max UTS}}{3} = 1066.67 \text{ MPa}$

For AISI 4337 steel

- Max UTS (MPa) = 1448
- Allowable static stress = $\frac{\text{Max UTS}}{3} = 482.67 \text{ MPa}$

Sample calculation for the pinion gear at input shaft

Pinion gear at input shaft is made using **High carbon steel**

$$\begin{aligned} \text{Applied tangential load on tooth} &= \frac{2200}{0.4101} \times 1.25 \\ &= 6705.11 \text{ N} \end{aligned}$$

$$C_v = \frac{3}{(3+0.4101)} = 0.8797$$

$$\begin{aligned} \text{Allowable tangential load on tooth} &= 1066.67 \times 0.8797 \times 31.25 \times \pi \times 2.5 \times 0.1084 \\ &= 24965.92155 \text{ N} \end{aligned}$$

So, allowable tangential load on tooth > Applied tangential load on tooth

So assumed module is suitable

	Input gear	Layshaft gear	1st gear		2nd gear		3rd gear		Reverse gear			
	1 (pinion)	2(gear)	3 (Pinion)	4(gear)	5 (gear)	6(Pinion)	7 (gear)	8(pinion)	9(reverse)	10(idle)	10 idle()	11 (gear)
Module	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	4	4	4	4
Material	High c.steel	AISI 4337 steel	High c steel	High c steel	AISI 4337 steel	AISI 4337 steel	AISI 4337 steel	AISI 4337 steel	High c steel	High c steel	High c steel	High c steel
Maximum UTS (MPa)	3200	1448	3200	3200	1448	1448	1448	1448	3200	3200	3200	1448
Allow. Static stress (MPa) (Max UTS/3)	1066.6667	482.66667	1066.6667	1066.6667	482.6666667	482.6666667	482.6666667	482.6666667	1066.666667	1066.667	1066.667	482.6666667
Condition	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut
Brinell hardness (min)	600	435	600	600	435	435	435	435	600	600	600	435
Young's modulus (GPa)	205	205	205	205	205	205	205	205	205	205	205	205
Flexural endurance limit (MPa)	1050	761.25	1050	1050	761.25	761.25	761.25	761.25	1050	1050	1050	761.25
Surface endurance limit	1610	1148	1610	1610	1148	1148	1148	1148	1610	1610	1610	1148
Nu. Of teeth	20	100	40	80	60	60	72	48	20	20	20	40
Module (mm)	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	4	4	4	4
P.C.D (mm)	50	250	100	200	150	150	180	120	80	80	80	160
RPM	156.66	31.33	31.33	15.665	31.33	31.33	31.33	46.995	31.33	31.33	31.33	15.665
Pitch line velocity (ms-1)	0.4101349	0.4101087	0.1640435	0.1640435	0.246065245	0.246065245	0.295278294	0.295278294	0.131234797	0.131235	0.131235	0.131234797
Velocity factor	0.8797306	0.8797373	0.9481538	0.9481538	0.924195841	0.924195841	0.910393518	0.910393518	0.958088484	0.958088	0.958088	0.958088484
Lewis tooth f f (y)	0.1084	0.14488	0.1312	0.1426	0.1388	0.1388	0.141333333	0.135	0.1084	0.1084	0.1084	0.1312
face width (mm)	31.25	31.25	31.25	31.25	31.25	31.25	31.25	31.25	50	50	50	50
Allow. Tangential load (with assumed module)	24965.922	15099.017	32567.264	35397.041	15196.39894	15196.39894	15242.66726	14559.6232	69605.48986	69605.49	69605.49	38121.20593
Transmitted Power	2200	2200	2200	2200	2200	2200	2200	2200	2200	2200	2200	2200
Service factor	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25
Applied tangential load	6705.1106	6705.5386	16763.847	16763.847	11175.8977	11175.8977	9313.24808	9313.24808	20954.80818	20954.81	20954.81	20954.80818
Applied tangential load (Without Service factor)	5364.0885	5364.4309	13411.077	13411.077	8940.718156	8940.718156	7450.598464	7450.598464	16763.84654	16763.85	16763.85	16763.84654
Module suitability	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable

Checking safety against tooth breakage

Calculating the dynamic load

$$W_D = W_T + W_I$$

W_D = Dynamic tooth load

W_T = Steady load due to transmitted torque (Without considering service factor)

W_I = Increment load due to dynamic action

$$W_D = W_T + \frac{21V(b.C + W_T)}{21V + \sqrt{b.C + W_T}}$$

V = Pitch line velocity

b = Face width of gears in mm

C = Deformation or dynamic factor in N/mm

Calculating tooth error

Our pitch line velocities are below 1.25 ms^{-1} and our pitches are 2.5 mm and 4mm

Table 28.6. Values of maximum allowable tooth error in action (e) verses pitch line velocity, for well cut commercial gears.

Pitch line velocity (v) m/s	Tooth error in action (e) mm	Pitch line velocity (v) m/s	Tooth error in action (e) mm	Pitch line velocity (v) m/s	Tooth error in action (e) mm
1.25	0.0925	8.75	0.0425	16.25	0.0200
2.5	0.0800	10	0.0375	17.5	0.0175
3.75	0.0700	11.25	0.0325	20	0.0150
5	0.0600	12.5	0.0300	22.5	0.0150
6.25	0.0525	13.75	0.0250	25 and over	0.0125
7.5	0.0475	15	0.0225		

Table 28.7. Values of tooth error in action (e) verses module.

Module (m) in mm	Tooth error in action (e) in mm		
	First class commercial gears	Carefully cut gears	Precision gears
Upto 4	0.051	0.025	0.0125
5	0.055	0.028	0.015
6	0.065	0.032	0.017
7	0.071	0.035	0.0186
8	0.078	0.0386	0.0198
9	0.085	0.042	0.021
10	0.089	0.0445	0.023
12	0.097	0.0487	0.0243
14	0.104	0.052	0.028
16	0.110	0.055	0.030
18	0.114	0.058	0.032
20	0.117	0.059	0.033

So we took tooth error as **0.04 mm**

And our all the pinions and gears are made from steel and has full depth involute system

So we took deformation factor as **456 N-mm**

Table 28.5. Values of deformation factor (C).

Material		Involute tooth form	Values of deformation factor (C) in N-mm				
Pinion	Gear		Tooth error in action (e) in mm				
			0.01	0.02	0.04	0.06	0.08
Cast iron	Cast iron	14½°	55	110	220	330	440
Steel	Cast iron		76	152	304	456	608
Steel	Steel		110	220	440	660	880
Cast iron	Cast iron	20° full depth	57	114	228	342	456
Steel	Cast iron		79	158	316	474	632
Steel	Steel		114	228	456	684	912
Cast iron	Cast iron	20° stub	59	118	236	354	472
Steel	Cast iron		81	162	324	486	648
Steel	Steel		119	238	476	714	952

Calculating the Static tooth load

$$W_S = \sigma_e \cdot b \cdot \pi \cdot m \cdot y$$

σ_e – Flexural Endurance limit

For steel $\sigma_e = 1.75 \cdot \text{Brinell}$

Sample calculation considering input pinion gear

$$W_T = \frac{2200}{0.4101} = 5364.08 \text{ N (Without considering service factor)}$$

$$W_I = \frac{21V(b.C+W_T)}{21V+\sqrt{b.C+W_T}} = \frac{21 \times 0.4101(31.25 \times 4.456 + 5364.08)}{21 \times 0.4101 + \sqrt{31.25 \times 4.456 + 5364.08}}$$
$$= 1136.3467 \text{ N}$$

$$\text{Dynamic tooth load } W_D = 5364.08 + 1136.3467$$

$$= 6500.43 \text{ N}$$

Input pinion gear is made of **high carbon steel**

Brinell value for High carbon steel = 600 (We consider the material with highest brinell)

Mechanical Properties	Metric
Hardness, Brinell	163 - 600
Hardness, Knoop	195 - 769
Hardness, Rockwell B	43.0 - 100
Hardness, Rockwell C	10.0 - 70.0
Hardness, Vickers	182 - 748
Tensile Strength, Ultimate	161 - 3200 MPa
Tensile Strength, Yield	275 - 3340 MPa
Elongation at Break	0.500 - 30.0 %
Reduction of Area	13.4 - 73.0 %
Modulus of Elasticity	13.8 - 235 GPa

$$\sigma_e = 1.75 \cdot 600 = 1050 \text{ MPa}$$

$$W_s = \sigma_e \cdot b \cdot \pi \cdot m \cdot y$$

$$= 1050 \times 31.25 \times \pi \times 2.5 \times 0.1084$$

$$= 27935.65 \text{ N}$$

$$\frac{W_s}{W_D} = 4.297$$

For Safety against tooth breakage at steady loads $W_s > 1.25W_D$

For Safety against tooth breakage at shock loads $W_s > 1.5W_D$

So the input pinion is safe from tooth breakage at both steady and shock loads

Checking the safety against premature wear of tooth

Calculating wear tooth load

$$W_w = D \cdot b \cdot Q \cdot K$$

W_w – Maximum or limiting load for wear in newtons

D – Pitch circle diameter

b – Face width

Q – Ratio factor

K – Load – stress factor

Sample calculation using input pinion gear

$$Q = \frac{2 \times V.R}{V.R + 1} \quad \text{For external gears}$$

$$= \frac{2 \times 100}{100 + 50} = 1.667$$

$$\text{Load stress factor } K = \frac{(\sigma_{es})^2 \cdot \sin \psi}{1.4} \left(\frac{1}{E_p} + \frac{1}{E_G} \right)$$

σ_{es} = Surface endurance limit in MPa or N/mm²

ψ = Pressure Angle

E_p = Young's modulus for the material of the pinion in N/mm²

E_G = Young's modulus for the material of the pinion in N/mm²

Both pinion and gear are made from steel (High carbon and AISI 4337)

Young's modulus of steel = 205 GPa

σ_{es} for steel = 2.8 × Brinell - 70

$$\begin{aligned} \sigma_{es} &= 2.8 \times 600 - 70 \quad (\text{For high carbon steel}) \\ &= 1610 \text{ MPa} \end{aligned}$$

$$\begin{aligned} K &= \frac{(\sigma_{es})^2 \cdot \sin \psi}{1.4} \left(\frac{1}{E_p} + \frac{1}{E_G} \right) \\ &= \frac{(1610)^2 \cdot \sin 20}{1.4} \left(\frac{1}{205} + \frac{1}{205} \right) \frac{1}{1000} \\ &= 6.17805 \end{aligned}$$

Wear tooth load

$$\begin{aligned} W_w &= D \cdot b \cdot Q \cdot K \\ &= 50 \times 31.25 \times 1.667 \times 6.17805 \\ &= 16088.6762 \text{ N} \end{aligned}$$

$$W_D = 6500.43 \text{ N}$$

$$\text{So, } W_w > W_D$$

So, Input pinion is **safe from premature wear**

	Input gear	Layshaft gear	1st gear		2nd gear		3rd gear		Reverse gear			
	1 (pinion)	2(gear)	3 (Pinion)	4(gear)	5 (gear)	6(Pinion)	7 (gear)	8(pinion)	9(reverse)	10(idle)	10 idle()	11 (gear)
Module	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	4	4	4	4
Material	High carbon steel	AISI 4337 steel	High c steel	High c steel	AISI 4337 steel	AISI 4337 steel	AISI 4337 steel	AISI 4337 steel	High c steel	High c steel	High c steel	High c steel
Maximum UTS (MPa)	3200	1448	3200	3200	1448	1448	1448	1448	3200	3200	3200	1448
Allow. Static stress (MPa) (Max UTS/3)	1066.6667	482.66667	1066.6667	1066.6667	482.6666667	482.6666667	482.6666667	482.6666667	1066.666667	1066.667	1066.667	482.6666667
Condition	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut	Ordinary cut
Brinell hardness (min)	600	435	600	600	435	435	435	435	600	600	600	435
Young's modulus (GPa)	205	205	205	205	205	205	205	205	205	205	205	205
Flexural endurance limit (MPa)	1050	761.25	1050	1050	761.25	761.25	761.25	761.25	1050	1050	1050	761.25
Surface endurance limit	1610	1148	1610	1610	1148	1148	1148	1148	1610	1610	1610	1148
Nu. Of teeth	20	100	40	80	60	60	72	48	20	20	20	40
Module (mm)	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	4	4	4	4
P.C.D (mm)	50	250	100	200	150	150	180	120	80	80	80	160
RPM	156.66	31.33	31.33	15.665	31.33	31.33	31.33	46.995	31.33	31.33	31.33	15.665
Pitch line velocity (ms-1)	0.4101349	0.4101087	0.1640435	0.1640435	0.246065245	0.246065245	0.295278294	0.295278294	0.131234797	0.131235	0.131235	0.131234797
Velocity factor	0.8797306	0.8797373	0.9481538	0.9481538	0.924195841	0.924195841	0.910393518	0.910393518	0.958088484	0.958088	0.958088	0.958088484
Lewis tooth f f (y)	0.1084	0.14488	0.1312	0.1426	0.1388	0.1388	0.141333333	0.135	0.1084	0.1084	0.1084	0.1312
face width (mm)	31.25	31.25	31.25	31.25	31.25	31.25	31.25	31.25	50	50	50	50
Allow. Tangential load (with assumed module)	24965.922	15099.017	32567.264	35397.041	15196.39894	15196.39894	15242.66726	14559.6232	69605.48986	69605.49	69605.49	38121.20593
Transmitted Power	2200	2200	2200	2200	2200	2200	2200	2200	2200	2200	2200	2200
Service factor	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25
Applied tangential load	6705.1106	6705.5386	16763.847	16763.847	11175.8977	11175.8977	9313.24808	9313.24808	20954.80818	20954.81	20954.81	20954.80818

Applied tangential load (Without Service factor)	5364.088 5	5364.430 9	13411.07 7	13411.07 7	8940.71815 6	8940.71815 6	7450.59846 4	7450.59846 4	16763.8465 4	16763.8 5	16763.8 5	16763.8465 4
Module suitability	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable	Suitable
Tooth error (mm)	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
Deformation factor (C) (N/mm)	456	456	456	456	456	456	456	456	456	456	456	456
Increment load (N)	1136.346 7	1136.288 9	561.3182 9	561.3182 9	761.087508 6	761.087508 6	876.556649 5	876.556649 5	540.681531 1	540.681 5	540.681 5	540.681531 1
Total dynamic load (Wd)	6500.435 2	6500.719 8	13972.39 6	13972.39 6	9701.80566 5	9701.80566 5	8327.15511 3	8327.15511 3	17304.5280 7	17304.5 3	17304.5 3	17304.5280 7
Static tooth load (Ws)	27935.63 1	27069.21 4	33811.39 1	36749.27 1	25933.2337 6	25933.2337 6	26406.5588 7	25223.2461 7	71515.2151 7	71515.2 2	71515.2 2	62753.9415 7
Ws/Wd	4.297501 6	4.164033 4	2.419870 7	2.630133 9	2.67303166 6	2.67303166 6	3.17113810 4	3.02903521 7	4.13274576 8	4.13274 6	4.13274 6	3.62644628 6
Safety of tooth breakage against steady loads	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe
Safety of tooth breakage against shock loads	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe
Ratio factor (Q)	1.666666 7	1.666667	1.333333 3	1.333333 3	1 1	1 1	1.2 1.2	1.2 1.2	1 1	1 1	1.33333 3	1.33333333 3
Load stress factor (K)	6.178051 7	3.141113	6.178051 7	6.178051 7	3.14111299 6	3.14111299 6	3.14111299 6	3.14111299 6	6.17805166 2	6.17805 2	6.17805 2	3.14111299 6
Wear tooth load	16088.67 6	40899.91 7	25741.88 2	51483.76 4	14723.9671 7	14723.9671 7	21202.5127 3	14135.0084 8	24712.2066 5	24712.2 1	32949.6 1	33505.2052 9
Safety against wear	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe	Safe

3.3.2 Bearing calculations

Our decided bearing placement in gearbox

Input shaft = 2 bearings at pulley end and gearbox housing end

Lay shaft = 2 bearing at gearbox housing

Idler shaft = One bearing at left side of gearbox housing

Output shaft = One bearing at left side of gearbox housing

Selected bearing type = Single row cylindrical roller bearing

Service factor for the roller bearing is assumed as 1.2

Dynamic load rating calculation

$$L = \left(\frac{C}{W} \right)^k \times 10^6$$

L= rating life

C= Basic dynamic load rating

W= Equivalant dynamic load

K =10/3 for roller bearings

N = speed in rpm

Relationship between life in revolutions (L) and life in working hours (L_H)

$$L = 60N \times L_H$$

Life of bearing in hours (L_H) is assumed as 50000 from below table according to our application

Table 27.3. Life of bearings for various types of machines.

<i>S. No.</i>	<i>Application of bearing</i>	<i>Life of bearing, in hours</i>
1.	Instruments and apparatus that are rarely used (a) Demonstration apparatus, mechanism for operating sliding doors (b) Aircraft engines	500 1000 – 2000
2.	Machines used for short periods or intermittently and whose breakdown would not have serious consequences <i>e.g.</i> hand tools, lifting tackle in workshops, and operated machines, agricultural machines, cranes in erecting shops, domestic machines.	4000 – 8000
3.	Machines working intermittently whose breakdown would have serious consequences <i>e.g.</i> auxiliary machinery in power stations, conveyor plant for flow production, lifts, cranes for piece goods, machine tools used frequently.	8000 – 12 000
4.	Machines working 8 hours per day and not always fully utilised <i>e.g.</i> stationary electric motors, general purpose gear units.	12 000 – 20 000
5.	Machines working 8 hours per day and fully utilised <i>e.g.</i> machines for the engineering industry, cranes for bulk goods, ventilating fans, counter shafts.	20 000 – 30 000
6.	Machines working 24 hours per day <i>e.g.</i> separators, compressors, pumps, mine hoists, naval vessels.	40 000 – 60 000
7.	Machines required to work with high degree of reliability 24 hours per day <i>e.g.</i> pulp and paper making machinery, public power plants, mine-pumps, water works.	100 000 – 200 000

Input shaft bearing calculation

Diameter of the shaft	30	mm
RPM	156.66	rpm
Life in revolutions	469980000	

Vertical and horizontal reactions at bearings are calculated from shear force diagram

Vertical shear force diagram

Reactions

Location: 85.25 (mm), Force Reaction = 1589.9135 (N)

Location: 480.62 (mm), Force Reaction = -8295.0241 (N)

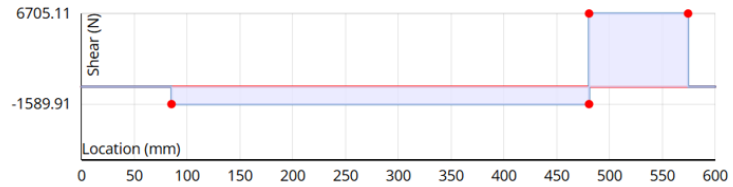
Shear Diagram

(Max +ve)Shear Load (N): 6705.111,

Location (mm): 480.620, 574.370

(Max -ve)Shear Load (N): -1589.914,

Location (mm): 85.250, 480.620



Horizontal shear force diagram

Reactions

Location: 85.25 (mm), Force Reaction = -1277.2198 (N)

Location: 480.62 (mm), Force Reaction = 3106.0398 (N)

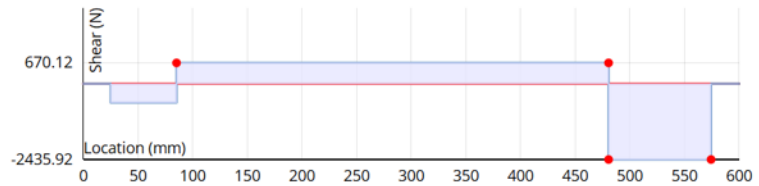
Shear Diagram

(Max +ve)Shear Load (N): 670.120,

Location (mm): 85.250, 480.620

(Max -ve)Shear Load (N): -2435.920,

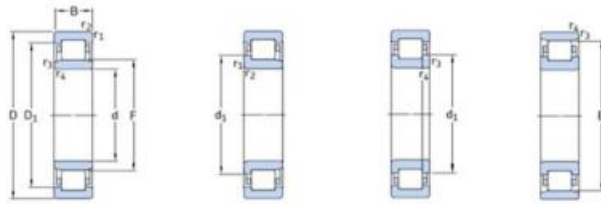
Location (mm): 480.620, 574.370



	Radial X	Radial Y	Radial load (N)	Axial (N) load	Resultant load (with S.F) (N)	Basic dynamic loading (kN)
Bearing LHS	1589.9135	1277.2198	2039.390927	0	2447.26911	33.73702641
Bearing RHS	8295.0241	3106.0398	8857.477522	0	10628.973	67.31550653

			Selected bearing
Maximum basic dynamic load rating on LHS bearing	33.737026	kN	N206 ECP
Maximum basic dynamic load rating on RHS bearing	67.315507	kN	NU 2306 ECP

Single row cylindrical roller bearings



NU			NJ			NUP			N	
Principal dimensions			Basic load		Fatigue load limit	Speed ratings		Mass	Designation	
			dynamic	static		Referen ce speed	Limiting speed		Bearing with standard cage	Alternative standard cage1)
d	D	B	C	C0	Pu					
	mm		KN			r/min	kg			
25	52	15	27.17	25.65	3,35	14 000	16 000	0,13	N 205 ECP	—
	52	18	32.395	32.3	4,25	14 000	16 000	0,16	NU 2205 ECP	ML
	52	18	32.395	32.3	4,25	14 000	16 000	0,17	NJ 2205 ECP	ML
	52	18	32.395	32.3	4,25	14 000	16 000	0,17	NUP 2205 ECP	ML
	62	17	44.175	34.675	4,55	12 000	15 000	0,23	* NU 305 ECP	J, ML
	62	17	44.175	34.675	4,55	12 000	15 000	0,24	* NJ 305 ECP	J, ML
	62	17	44.175	34.675	4,55	12 000	15 000	0,25	* NUP 305 ECP	J, ML
	62	17	44.175	34.675	4,55	12 000	15 000	0,24	* N 305 ECP	—
	62	24	60.8	52.25	6,95	12 000	15 000	0,34	* NU 2305 ECP	J, ML
	62	24	60.8	52.25	6,95	12 000	15 000	0,35	* NJ 2305 ECP	J, ML
	62	24	60.8	52.25	6,95	12 000	15 000	0,36	* NUP 2305 ECP	J, ML
30	55	13	17.005	16.435	1,86	15 000	15 000	0,12	NU 1006	—
	62	16	41.8	34.675	4,5	13 000	14 000	0,2	* NU 206 ECP	J, ML, PH
	62	16	41.8	34.675	4,5	13 000	14 000	0,21	* NJ 206 ECP	J, ML, PH
	62	16	41.8	34.675	4,5	13 000	14 000	0,21	* NUP 206 ECP	J, ML, PH
	62	16	41.8	34.675	4,5	13 000	14 000	0,2	* N 206 ECP	—
	62	20	52.25	46.55	6,1	13 000	14 000	0,26	* NU 2206 ECP	J, ML, PH
	62	20	52.25	46.55	6,1	13 000	14 000	0,26	* NJ 2206 ECP	J, ML, PH
	62	20	52.25	46.55	6,1	13 000	14 000	0,27	* NUP 2206 ECP	J, ML, PH
	72	19	55.575	45.6	6,2	11 000	12 000	0,36	* NU 306 ECP	J, M, ML
	72	19	55.575	45.6	6,2	11 000	12 000	0,37	* NJ 306 ECP	J, M, ML
	72	19	55.575	45.6	6,2	11 000	12 000	0,38	* NUP 306 ECP	J, M, ML
	72	19	55.575	45.6	6,2	11 000	12 000	0,36	* N 306 ECP	—
	72	27	78.85	71.25	9,65	11 000	12 000	0,53	* NU 2306 ECP	ML
	72	27	78.85	71.25	9,65	11 000	12 000	0,54	* NJ 2306 ECP	ML
	72	27	78.85	71.25	9,65	11 000	12 000	0,55	* NUP 2306 ECP	ML
	90	23	57.475	50.35	6,8	9 000	11 000	0,75	NU 406	MA
90	23	57.475	50.35	6,8	9 000	11 000	0,79	NJ 406	MA	
35	62	14	34.01	36.1	4,55	13 000	13 000	0,16	NU 1007 ECP	PH

Layshaft bearing calculation

Diameter of the shaft (mm)	55
RPM	31.33
Life in revolutions	93990000

1st gear

Vertical and horizontal Reactions

Reactions

Location: 0 (mm), Force Reaction = -6896.7694 (N)

Location: 759.38 (mm), Force Reaction = -1895.2644 (N)

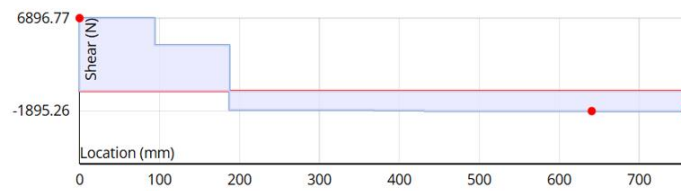
Shear Diagram

(Max +ve)Shear Load (N): 6896.769,

Location (mm): 0.000

(Max -ve)Shear Load (N): -1895.264,

Location (mm): 640.630, 759.380



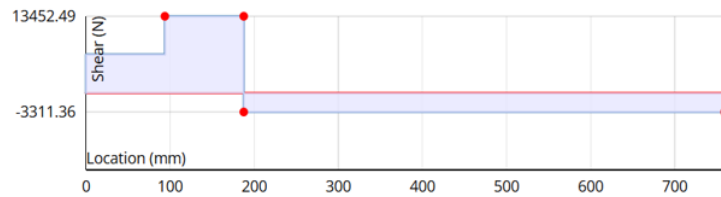
Shear Diagram

(Max +ve)Shear Load (N): 13452.491,

Location (mm): 93.750, 187.500

(Max -ve)Shear Load (N): -3311.355,

Location (mm): 187.500, 759.380



	Radial X	Radial Y	Radial load	Axial load	Resultant load (with S.F)	Basic dynamic loading (kN)
Bearing LHS	6896.7694	6511.3251	9484.871	0	11381.8455	44.4771736
Bearing RHS	1895.2644	3311.36	3815.381	0	4578.4572	17.89137234

2nd gear

Point Load

Reactions

Location: 0 (mm), Force Reaction = -4394.22 (N)

Location: 759.38 (mm), Force Reaction = -2363.9668 (N)

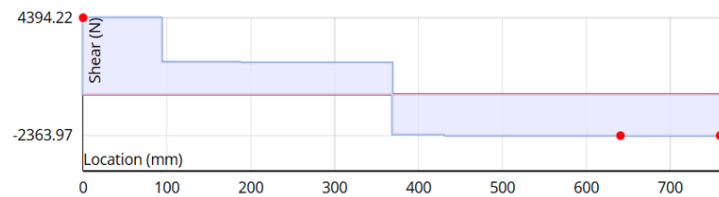
Shear Diagram

(Max +ve)Shear Load (N): 4394.220,

Location (mm): 0.000

(Max -ve)Shear Load (N): -2363.967,

Location (mm): 640.630, 759.380



Reactions

Location: 0 (mm), Force Reaction = 128.7455 (N)

Location: 759.38 (mm), Force Reaction = -4599.1046 (N)

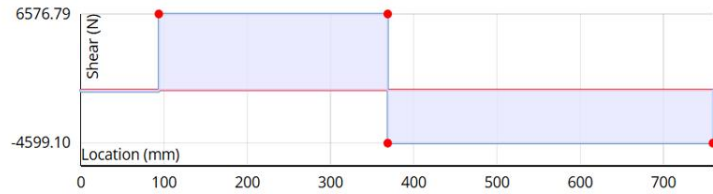
Shear Diagram

(Max +ve)Shear Load (N): 6576.793,

Location (mm): 93.750, 368.750

(Max -ve)Shear Load (N): -4599.105,

Location (mm): 368.750, 759.380



	Radial X	Radial Y	Radial load	Axial load	Resultant load (with S.F)	Basic dynamic loading (kN)
Bearing LHS	4394.22	128.7455	4396.106	0	5275.32677	20.61455011
Bearing RHS	2363.9668	4599.1046	5171.083	0	6205.29992	24.24863364

3rd gear

Reactions

Location: 0 (mm), Force Reaction = -3766.4891 (N)

Location: 759.38 (mm), Force Reaction = -2313.7487 (N)

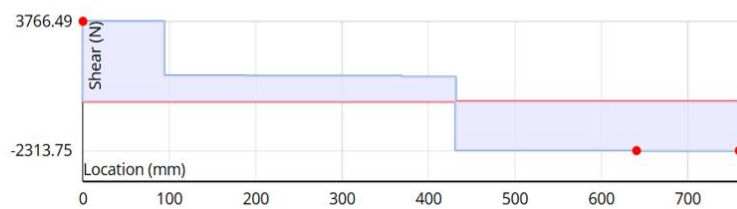
Shear Diagram

(Max +ve)Shear Load (N): 3766.489,

Location (mm): 0.000

(Max -ve)Shear Load (N): -2313.749,

Location (mm): 640.630, 759.380



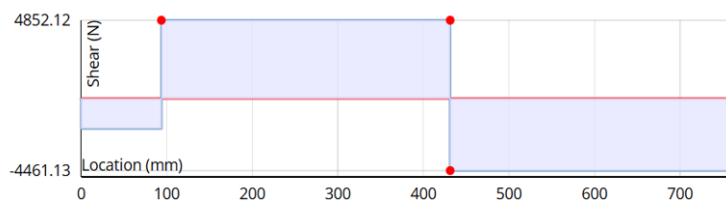
Shear Diagram

(Max +ve)Shear Load (N): 4852.117,

Location (mm): 93.750, 431.250

(Max -ve)Shear Load (N): -4461.131,

Location (mm): 431.250, 759.380



	Radial X	Radial Y	Radial load	Axial load	Resultant load (with S.F)	Basic dynamic loading
Bearing LHS	3766.4891	1951.4321	4241.996	0	5090.394821	19.89188608
Bearing RHS	2313.7487	4461.13	5025.447	0	6030.535967	23.56570338

Reverse gear

Reactions

Location: 0 (mm), Force Reaction = -5633.5232 (N)

Location: 759.38 (mm), Force Reaction = -18362.7839 (N)

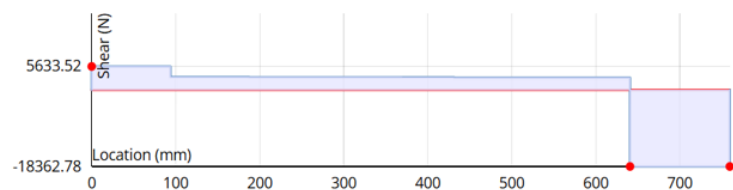
Shear Diagram

(Max +ve)Shear Load (N): 5633.523,

Location (mm): 0.000

(Max -ve)Shear Load (N): -18362.784,

Location (mm): 640.630, 759.380



Reactions

Location: 0 (mm), Force Reaction = 2947.4959 (N)

Location: 759.38 (mm), Force Reaction = -14979.9633 (N)

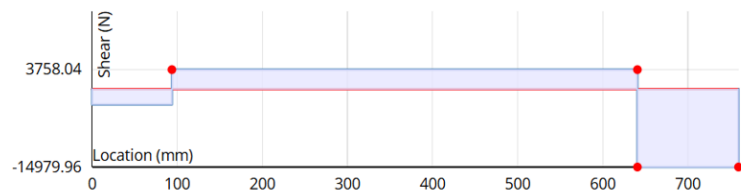
Shear Diagram

(Max +ve)Shear Load (N): 3758.043,

Location (mm): 93.750, 640.630

(Max -ve)Shear Load (N): -14979.963,

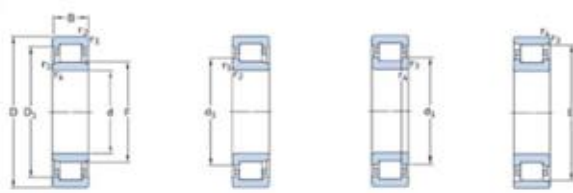
Location (mm): 640.630, 759.380



	Radial X	Radial Y	Radial load	Axial load	Resultant load (with S.F)	Basic dynamic loading
Bearing LHS	5633.5232	2947.4959	6358.0119	0	7629.6143	29.81446905
Bearing RHS	18362.7839	14979.963	23697.914	0	28437.497	111.126046

			Selected bearing
Maximum basic dynamic load rating on LHS bearing	29.8144691	kN	NU 1011 ECP
Maximum basic dynamic load rating on RHS bearing	111.126046	kN	NU 311 ECP

Single row cylindrical roller bearings



Principal dimensions	NU		NJ		NUP		N			
	Basic load		Fatigue load limit	Speed ratings		Mass	Designation			
	dynamic	static		Reference speed	Limiting speed		Bearing with standard cage	Alternative standard cage1)		
d	D	B	C	C0	Pu					
mm			KN			r/min	kg			
50	90	23	85.5	83.6	11,4	8 500	9 000	0,59	NUP 2210 ECP	J, M, ML, PH
	110	27	120.65	106.4	15	6 700	8 000	1,15	* NU 310 ECP	J, M, ML, PH
	110	27	120.65	106.4	15	6 700	8 000	1,15	* NJ 310 ECP	J, M, ML, PH
	110	27	120.65	106.4	15	6 700	8 000	1,15	* NUP 310 ECP	J, M, ML, PH
	110	27	120.65	106.4	15	6 700	8 000	1,15	* N 310 ECP	—
	110	40	176.7	176.7	24,5	6 700	8 000	1,75	* NU 2310 ECP	J, ML, PH
	110	40	176.7	176.7	24,5	6 700	8 000	1,75	* NJ 2310 ECP	J, ML, PH
	110	40	176.7	176.7	24,5	6 700	8 000	1,75	* NUP 2310 ECP	J, ML, PH
	130	31	123.5	120.65	16,6	6 000	7 000	2	NU 410	—
	130	31	123.5	120.65	16,6	6 000	7 000	2,05	NJ 410	—
	130	31	123.5	120.65	16,6	6 000	7 000	2,05	NUP 410	—
	130	31	123.5	120.65	16,6	6 000	7 000	2,05	N 410	—
55	90	18	54.34	56.025	8,3	8 500	8 500	0,39	NU 1011 ECP	ML
	100	21	91.675	90.25	12,2	7 500	8 000	0,66	* NU 211 ECP	J, M, ML
	100	21	91.675	90.25	12,2	7 500	8 000	0,67	* NJ 211 ECP	J, M, ML
	100	21	91.675	90.25	12,2	7 500	8 000	0,68	* NUP 211 ECP	J, M, ML
	100	21	91.675	90.25	12,2	7 500	8 000	0,65	* N 211 ECP	M
	100	25	108.3	112.1	15,3	7 500	8 000	0,79	* NU 2211 ECP	J, M, ML, PH
	100	25	108.3	112.1	15,3	7 500	8 000	0,81	* NJ 2211 ECP	J, M, ML, PH
	100	25	108.3	112.1	15,3	7 500	8 000	0,82	* NUP 2211 ECP	J, M, ML, PH
	100	25	108.3	112.1	15,3	7 500	8 000	0,82	* N 2211 ECP	M
	120	29	148.2	135.85	18,6	6 000	7 000	1,45	* NU 311 ECP	J, M, ML
	120	29	148.2	135.85	18,6	6 000	7 000	1,5	* NJ 311 ECP	J, M, ML
	120	29	148.2	135.85	18,6	6 000	7 000	1,5	* NUP 311 ECP	J, M, ML
120	29	148.2	135.85	18,6	6 000	7 000	1,45	* N 311 ECP	M	
120	43	220.4	220.4	30,5	6 000	7 000	2,2	* NU 2311 ECP	J, ML, PH	
120	43	220.4	220.4	30,5	6 000	7 000	2,25	* NJ 2311 ECP	J, ML, PH	
120	43	220.4	220.4	30,5	6 000	7 000	2,3	* NUP 2311 ECP	J, ML, PH	
140	33	134.9	133	18,6	5 600	6 300	2,5	NU 411	—	
140	33	134.9	133	18,6	5 600	6 300	2,55	NJ 411	—	
140	33	134.9	133	18,6	5 600	6 300	2,55	NUP 411	—	
140	33	134.9	133	18,6	5 600	6 300	2,55	N 411	—	
60	95	18	35.53	41.8	5,3	8 000	13 000	0,5	NU 1012 ML	—

Idle shaft bearing calculation

Diameter of the shaft	55	mm
RPM	31.33	rpm
Life in revolutions	93990000	

Vertical and horizontal reactions at bearings are calculated from shear force diagram

Reactions

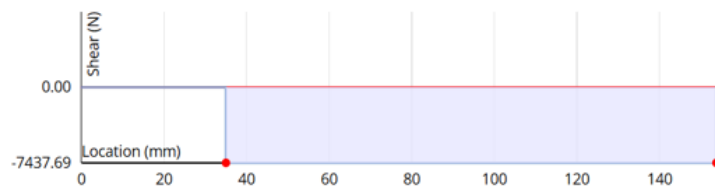
Location: 153.75 (mm), Force Reaction = -7437.6885 (N)

Location: 153.75 (mm), Moment Reaction = 883225.5046 (N-mm)

Shear Diagram

(Max -ve)Shear Load (N): -7437.688,

Location (mm): 35.000, 153.750



Reactions

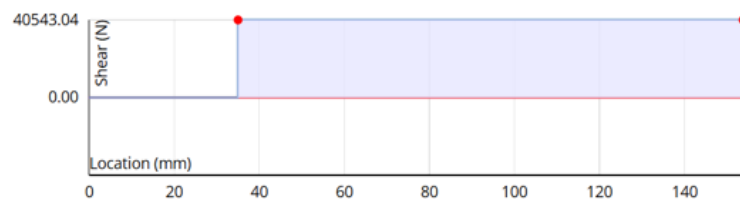
Location: 153.75 (mm), Force Reaction = 40543.0367 (N)

Location: 153.75 (mm), Moment Reaction = -4814485.6117 (N-mm)

Shear Diagram

(Max +ve)Shear Load (N): 40543.037,

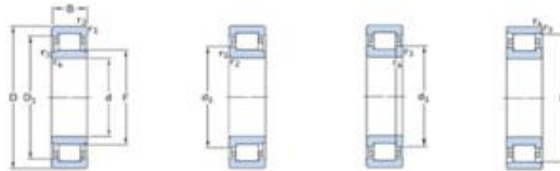
Location (mm): 35.000, 153.750



	Radial X	Radial Y	Radial load	Axial load	Resultant load (with S.F)	Basic dynamic loading	
Bearing LHS							NO
Bearing RHS	7437.689	40543.037	41219.62	0	49463.54345	193.2901486	kN

			Selected bearing
Maximum basic dynamic load rating on RHS bearing	193.29015	kN	NU 2311 ECP

Single row cylindrical roller bearings



		NU		NJ		NUP		N	
Principal dimensions		Basic load		Fatigue load limit	Speed ratings		Mass	Designation	
		dynamic	static		Reference speed	Limiting speed		Bearing with standard cage	Alternative standard cage1)
d	D	B	C	C0	P0	r/min	kg		
50	90	23	85.5	83.6	11.4	8 500	9 000	0.59	NUP 2210 ECP J, M, ML, PH
	110	27	120.65	106.4	15	6 700	8 000	1.15	* NU 310 ECP J, M, ML, PH
	110	27	120.65	106.4	15	6 700	8 000	1.15	* NJ 310 ECP J, M, ML, PH
	110	27	120.65	106.4	15	6 700	8 000	1.15	* NUP 310 ECP J, M, ML, PH
	110	27	120.65	106.4	15	6 700	8 000	1.15	* N 310 ECP -
	110	40	176.7	176.7	24.5	6 700	8 000	1.75	* NU 2310 ECP J, ML, PH
	110	40	176.7	176.7	24.5	6 700	8 000	1.75	* NJ 2310 ECP J, ML, PH
	110	40	176.7	176.7	24.5	6 700	8 000	1.75	NUP 2310 ECP J, ML, PH
	130	31	123.5	120.65	16.6	6 000	7 000	2	NU 410 -
	130	31	123.5	120.65	16.6	6 000	7 000	2.05	NJ 410 -
55	90	18	54.34	66.025	8.3	8 500	8 500	0.39	NU 1011 ECP ML
	90	18	54.34	66.025	8.3	8 500	8 500	0.42	NJ 1011 ECP ML
	100	21	91.675	90.25	12.2	7 500	8 000	0.66	* NU 211 ECP J, M, ML
	100	21	91.675	90.25	12.2	7 500	8 000	0.67	* NJ 211 ECP J, M, ML
	100	21	91.675	90.25	12.2	7 500	8 000	0.68	* NUP 211 ECP J, M, ML
	100	21	91.675	90.25	12.2	7 500	8 000	0.65	* N 211 ECP M
	100	25	108.3	112.1	15.3	7 500	8 000	0.79	* NU 2211 ECP J, M, ML, PH
	100	25	108.3	112.1	15.3	7 500	8 000	0.81	* NJ 2211 ECP J, M, ML, PH
	100	25	108.3	112.1	15.3	7 500	8 000	0.82	* NUP 2211 ECP J, M, ML, PH
	120	29	148.2	135.85	18.6	6 000	7 000	1.45	* NU 311 ECP J, M, ML
	120	29	148.2	135.85	18.6	6 000	7 000	1.5	* NJ 311 ECP J, M, ML
	120	29	148.2	135.85	18.6	6 000	7 000	1.5	* NUP 311 ECP J, M, ML
	120	30	148.2	135.85	18.6	6 000	7 000	1.45	* N 311 ECP M
	120	43	220.4	220.4	30.5	6 000	7 000	2.2	* NU 2311 ECP J, ML, PH
	120	43	220.4	220.4	30.5	6 000	7 000	2.25	NJ 2311 ECP J, ML, PH
	120	43	220.4	220.4	30.5	6 000	7 000	2.3	NUP 2311 ECP J, ML, PH
	140	33	134.9	133	18.6	5 600	6 300	2.5	NU 411 -
	140	33	134.9	133	18.6	5 600	6 300	2.55	NJ 411 -
60	95	18	35.53	41.8	5.3	8 000	13 000	0.5	NJ 1012 ML -

Output shaft bearing calculation

Diameter of shaft	65	mm
-------------------	----	----

1st gear	
RPM	15.665
Life in revolutions	46995000

Reactions

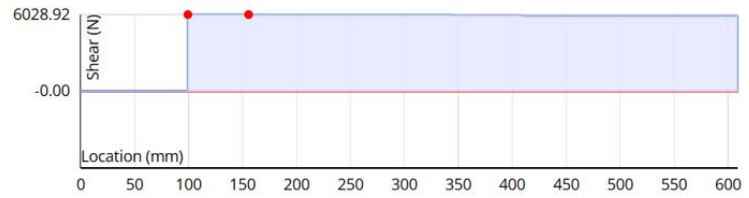
Location: 608.75 (mm), Force Reaction = 5887.5742 (N)

Location: 608.75 (mm), Moment Reaction = -3030687.9166 (N-mm)

Shear Diagram

(Max +ve)Shear Load (N): 6028.924,

Location (mm): 99.370, 155.620



Reactions

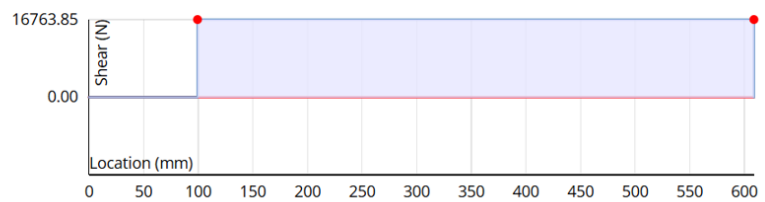
Location: 608.75 (mm), Force Reaction = 16763.8465 (N)

Location: 608.75 (mm), Moment Reaction = -8539168.1505 (N-mm)

Shear Diagram

(Max +ve)Shear Load (N): 16763.847,

Location (mm): 99.370, 608.750



	Radial X	Radial Y	Radial load	Axial load	Resultant load (with S.F)	Basic dynamic loading	
Bearing LHS							NO
Bearing RHS	5887.5742	16763.847	17767.669	0	21321.2034	67.67483561	kN

2nd gear	
RPM	31.33
Life in revolutions	93990000

Reactions

Location: 608.75 (mm), Force Reaction = 3853.7272 (N)

Location: 608.75 (mm), Moment Reaction = -1765879.1126 (N-mm)

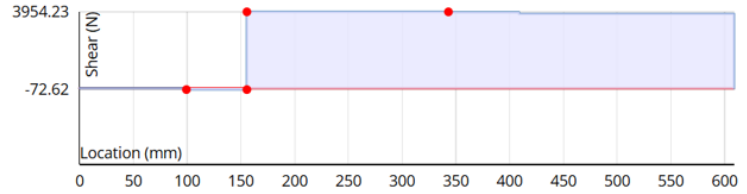
Shear Diagram

(Max +ve)Shear Load (N): 3954.230,

Location (mm): 155.620, 343.120

(Max -ve)Shear Load (N): -72.617,

Location (mm): 99.370, 155.620



Reactions

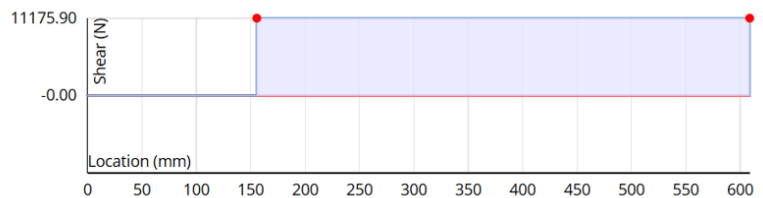
Location: 608.75 (mm), Force Reaction = 11175.8977 (N)

Location: 608.75 (mm), Moment Reaction = -5064134.5248 (N-mm)

Shear Diagram

(Max +ve)Shear Load (N): 11175.898,

Location (mm): 155.620, 608.750



	Radial X	Radial Y	Radial load	Axial load	Resultant load (with S.F)	Basic dynamic loading	
Bearing LHS							NO
Bearing RHS	3853.7272	11175.898	11821.671	0	14186.005	55.4350707	kN

3rd gear	
RPM	46.995
Life in revolutions	140985000

Reactions

Location: 608.75 (mm), Force Reaction = 3175.7781 (N)

Location: 608.75 (mm), Moment Reaction = -823102.8709 (N-mm)

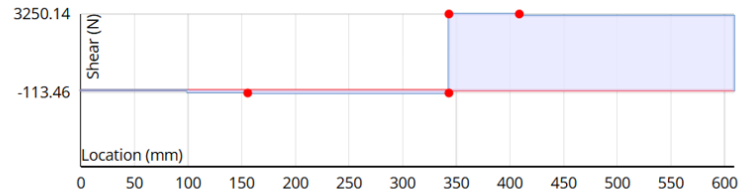
Shear Diagram

(Max +ve)Shear Load (N): 3250.138,

Location (mm): 343.120, 408.750

(Max -ve)Shear Load (N): -113.465,

Location (mm): 155.620, 343.120



Reactions

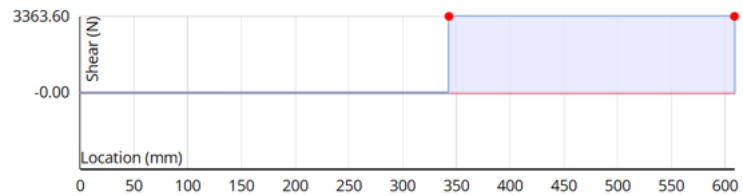
Location: 608.75 (mm), Force Reaction = 3363.6028 (N)

Location: 608.75 (mm), Moment Reaction = -893473.8245 (N-mm)

Shear Diagram

(Max +ve)Shear Load (N): 3363.603,

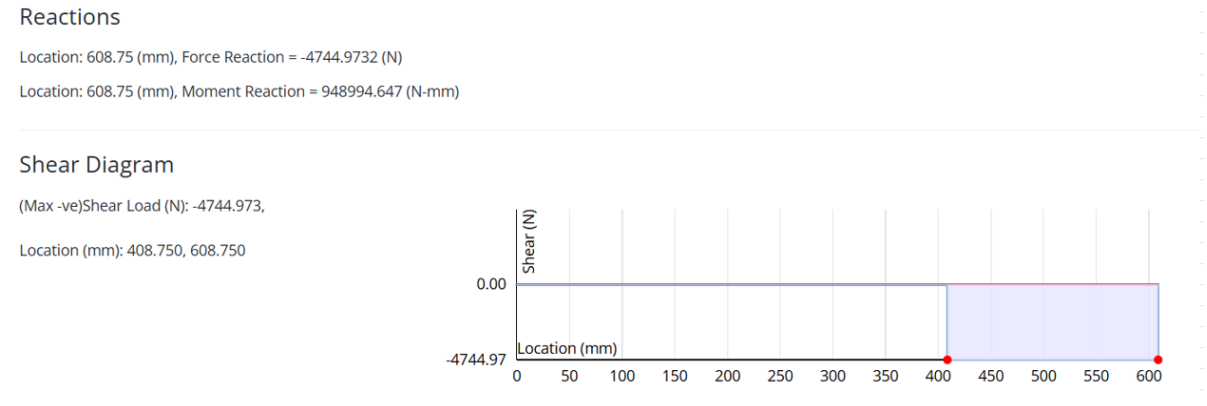
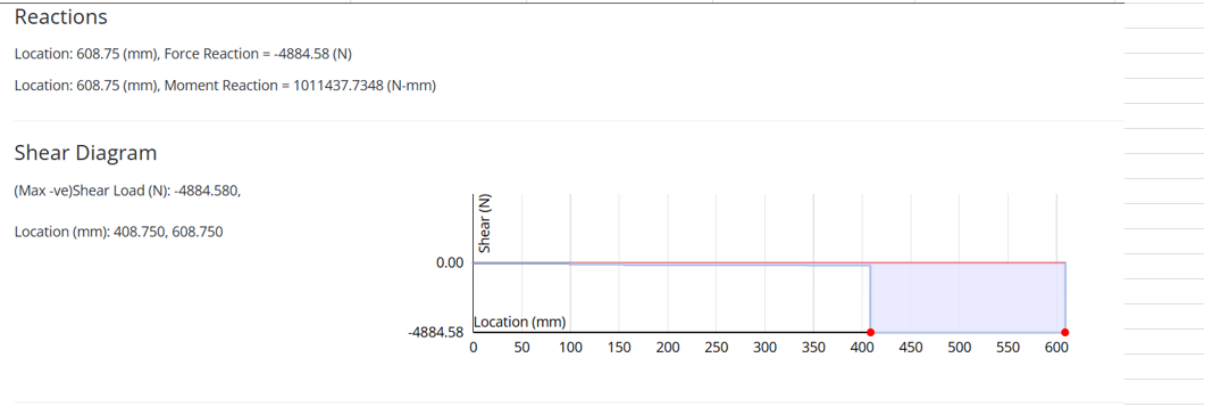
Location (mm): 343.120, 608.750



	Radial X	Radial Y	Radial load	Axial load	Resultant load (with S.F)	Basic dynamic loading	
Bearing LHS							NO
Bearing RHS	3175.7781	3363.6028	4625.9475	0	5551.13701	24.49817987	kN

Reverse gear

RPM	15.665
Life in revolutions	46995000



	Radial X	Radial Y	Radial load	Axial load	Resultant load (with S.F)	Basic dynamic loading	
Bearing LHS							NO
Bearing RHS	4884.58	4744.973	6809.8378	0	8171.80533	25.93782218	kN

			Selected bearing
--	--	--	------------------

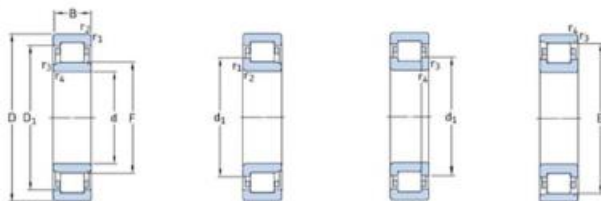
Maximum basic dynamic load rating on RHS bearing

67.674836

kN

NU 213 ECP

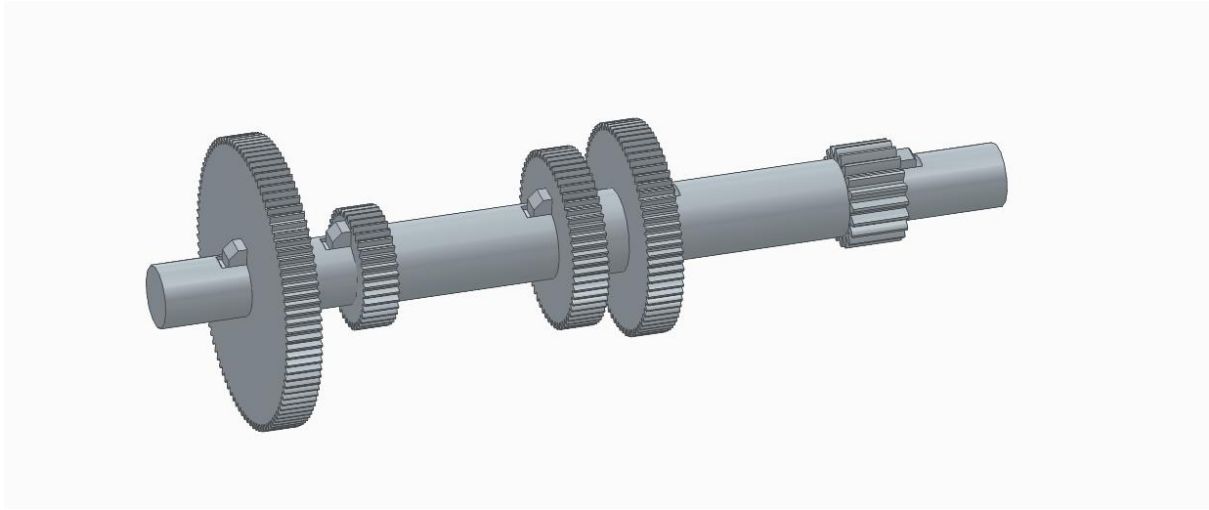
Single row cylindrical roller bearings



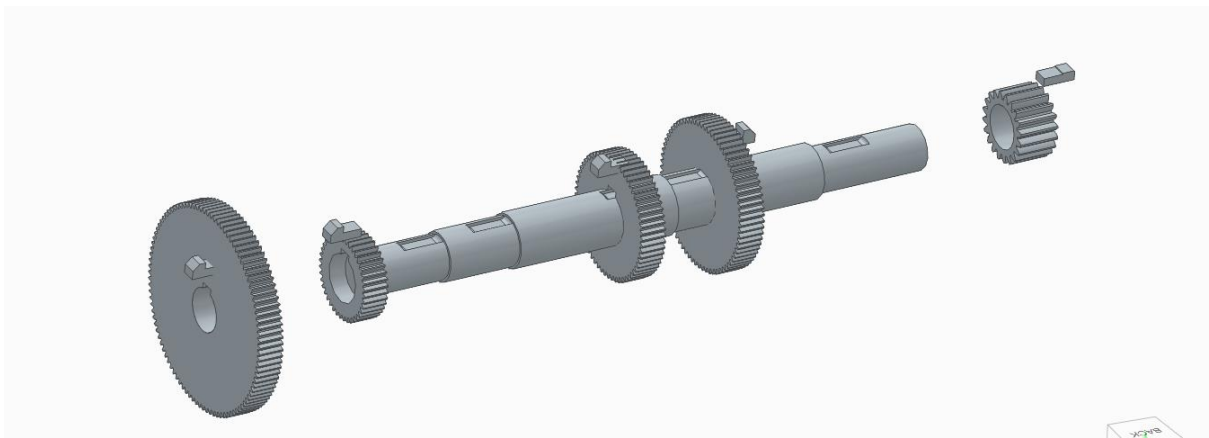
NU			NJ			NUP			N	
Principal dimensions			Basic load		Fatigue load limit	Speed ratings		Mass	Designation	
			dynamic	static		Referen ce speed	Limiting speed		Bearing with standard cage	Alternative standard cage1)
d	D	B	C	C0	Pu					
mm			KN			r/min		kg		
60	110	22	102.6	96.9	13,4	6 700	7 500	0,8	* NU 212 ECP	J, M, ML
	110	22	102.6	96.9	13,4	6 700	7 500	0,82	* NJ 212 ECP	J, M, ML
	110	22	102.6	96.9	13,4	6 700	7 500	0,86	* NUP 212 ECP	J, M, ML
	110	22	102.6	96.9	13,4	6 700	7 500	0,81	* N 212 ECP	M
	110	28	138.7	145.35	20	6 700	7 500	1,05	* NU 2212 ECP	J, M, ML
	110	28	138.7	145.35	20	6 700	7 500	1,1	* NJ 2212 ECP	J, M, ML
	110	28	138.7	145.35	20	6 700	7 500	1,1	* NUP 2212 ECP	J, M, ML
	130	31	164.35	152	21,2	5 600	6 700	1,75	* NU 312 ECP	J, M, ML
	130	31	164.35	152	21,2	5 600	6 700	1,85	* NJ 312 ECP	J, M, ML
	130	31	164.35	152	21,2	5 600	6 700	1,9	* NUP 312 ECP	J, M, ML
	130	31	164.35	152	21,2	5 600	6 700	1,75	* N 312 ECP	J, M
	130	46	247	251.75	34,5	5 600	6 700	2,75	* NU 2312 ECP	J, M, ML
	130	46	247	251.75	34,5	5 600	6 700	2,8	* NJ 2312 ECP	J, M, ML
	130	46	247	251.75	34,5	5 600	6 700	2,85	* NUP 2312 ECP	J, M, ML
	150	35	159.6	164.35	22	5 000	6 000	3	NU 412	—
	150	35	159.6	164.35	22	5 000	6 000	3,1	NJ 412	—
65	100	18	59.565	77.425	9,8	7 500	7 500	0,45	NU 1013 ECP	—
	120	23	115.9	112.1	15,6	6 300	6 700	1,05	* NU 213 ECP	J, M, ML
	120	23	115.9	112.1	15,6	6 300	6 700	1,05	* NJ 213 ECP	J, M, ML
	120	23	115.9	112.1	15,6	6 300	6 700	1,1	* NUP 213 ECP	J, M, ML
	120	23	115.9	112.1	15,6	6 300	6 700	1,05	* N 213 ECP	—
	120	31	161.5	171	24	6 300	6 700	1,4	* NU 2213 ECP	J, ML
	120	31	161.5	171	24	6 300	6 700	1,45	* NJ 2213 ECP	J, ML
	120	31	161.5	171	24	6 300	6 700	1,5	* NUP 2213 ECP	J, ML
	140	33	201.4	186.2	25,5	5 300	6 000	2,2	* NU 313 ECP	J, M, ML, PH
	140	33	201.4	186.2	25,5	5 300	6 000	2,3	* NJ 313 ECP	J, M, ML, PH
	140	33	201.4	186.2	25,5	5 300	6 000	2,35	* NUP 313 ECP	J, M, ML, PH
	140	33	201.4	186.2	25,5	5 300	6 000	2,2	* N 313 ECP	M
	140	48	270.75	275.5	38	5 300	6 000	3,2	* NU 2313 ECP	ML, PH

3.4 Final design

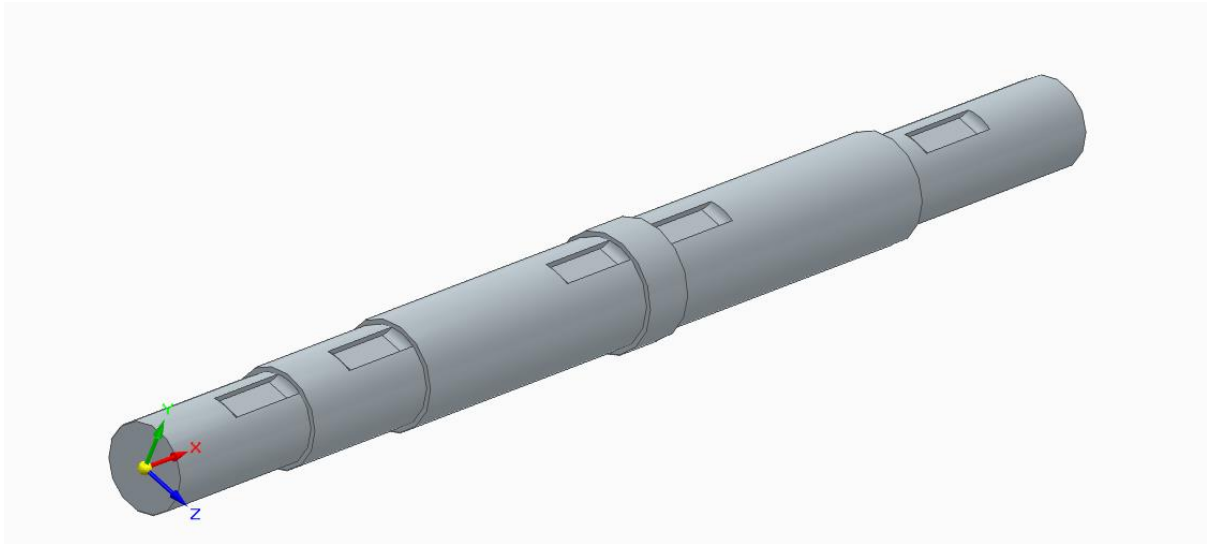
Lay shaft, lay shaft gears, keys and assembly



Lay shaft assembly



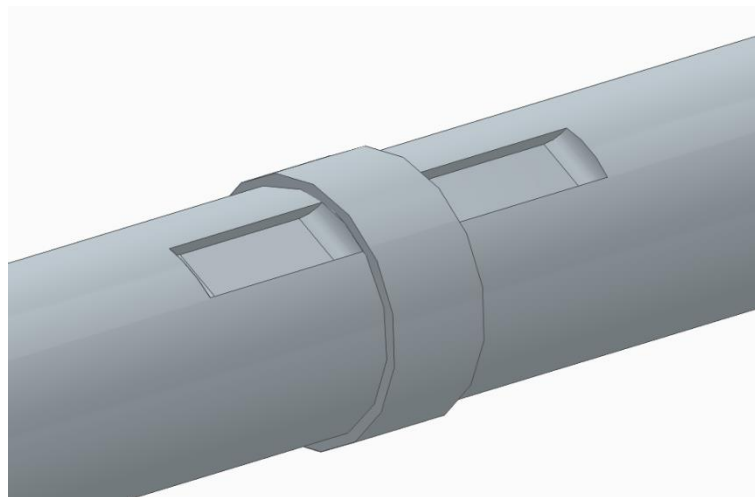
Exploded view



Lay shaft

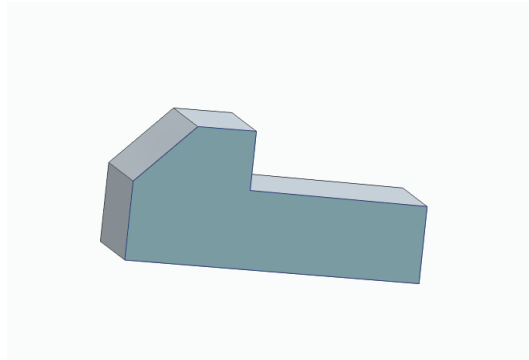
Lay shaft is used to transmit the power from input shaft to the output shaft. We designed lay shaft as stepped shaft to minimize the axial motion of the lay shaft gears. When designing the steps, we ensured that the diameters at all the locations exceeded the calculated minimum diameter required for the lay shaft. And also, we used rectangular gib head keys to secure the gear wheels on the lay shaft. Two cylindrical single row roller bearings were used to attach the layshaft to the housing and maintain the axial alignment.

Keyways



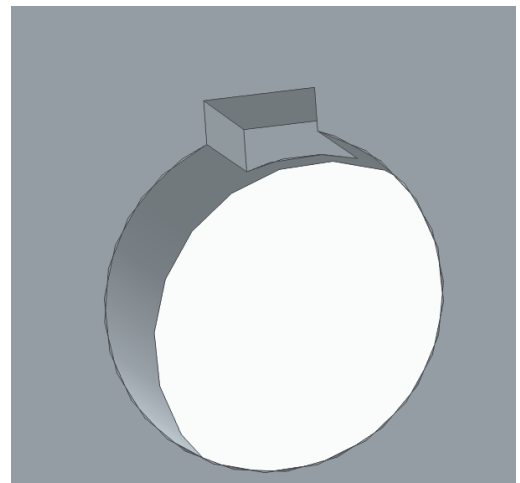
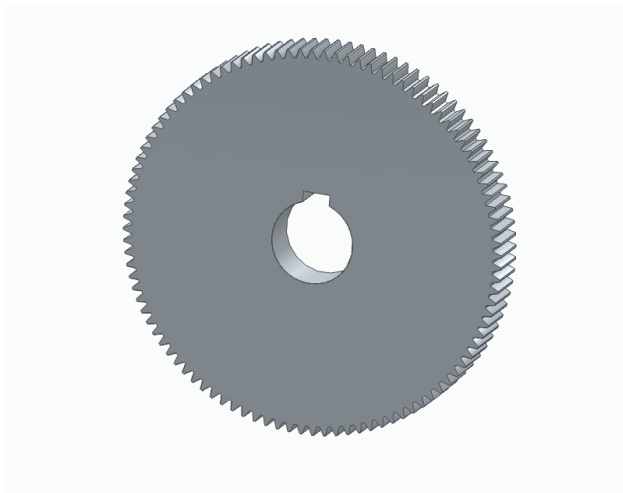
We designed the keyways of the layshaft as in above figure to facilitate face milling (up milling) operation when cut the keyway.

Key



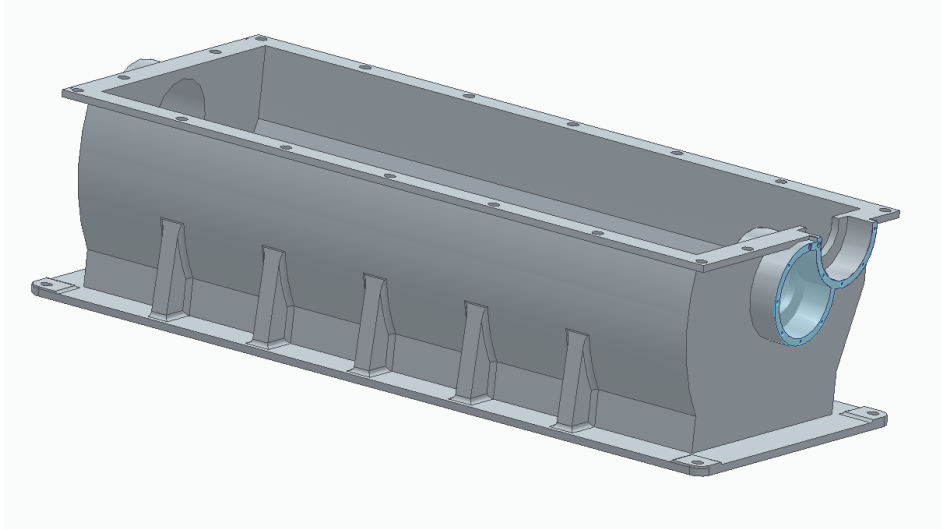
We used rectangular gib head key to attach gears to the lay shaft, with the head also prevents the axial motion of the gear wheel and also we used 1:100 taper on the key and keyway to maintain a tight fit between shaft and gear wheel.

Gears and keyways of the gears

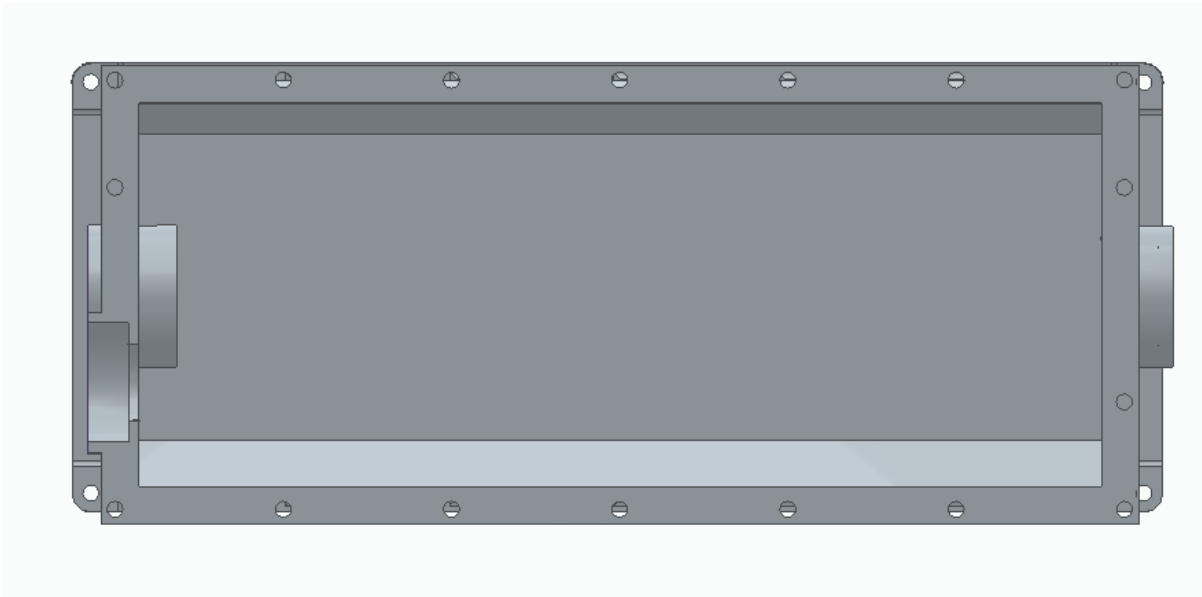


In the keyway we also added 1:100 taper to ensure the tightness between the shaft and the wheel

Bottom part of the housing

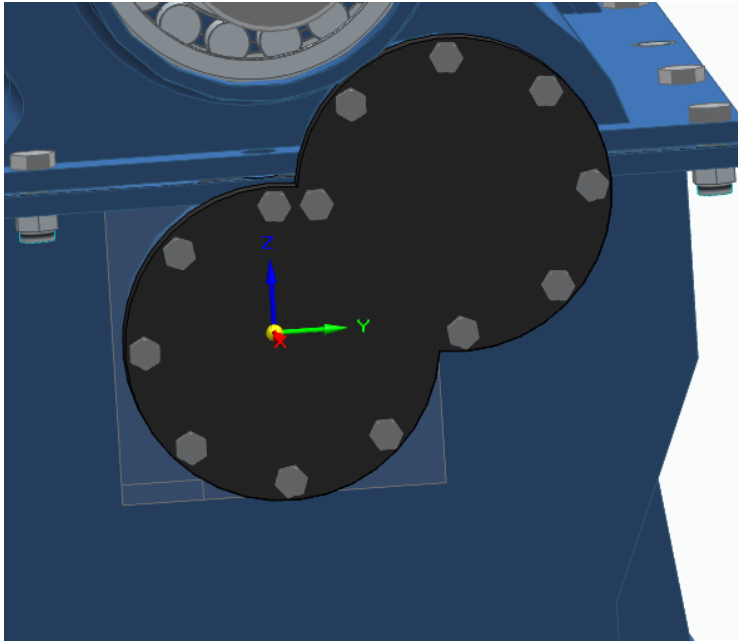


Bottom part of the housing



Inside view of the bottom part of the housing

The housing of the gearbox was made of cast iron with a thickness of 8 mm. This thickness was determined based on bending calculations performed at the two primary loaded surfaces. And also, there is a enough gap between the gears and the housing to ensure the safety of the gear wheels. We added ribs to support the casing on both sides. We used gasket between the bottom and upper part of the housing as a sealant to reduce the leakages. The holes of the housing relevant to lay shaft and idle shaft were designed as above manner to avoid the collision between two bearings of lay shaft and idle shaft. The base of the housing is produced with bolt holes to mount on a surface. We used M6 hexagonal head screws to attach the bearing caps used M12 hex head to connect the upper and lower part of the housings.



Bearing caps at the layshaft and idle shaft end

Bearing caps at the layshaft and idle shaft end also made in above manner to avoid the collision between layshaft and idle shaft bearings

3.5 Future works

We can do a topology optimization on the gears and reduce and optimize the weight of the gears.

We can do a FEA simulation on the gearbox housing and optimize the shape of the casing and increase the compactness and strength of the casing.

3.6 References

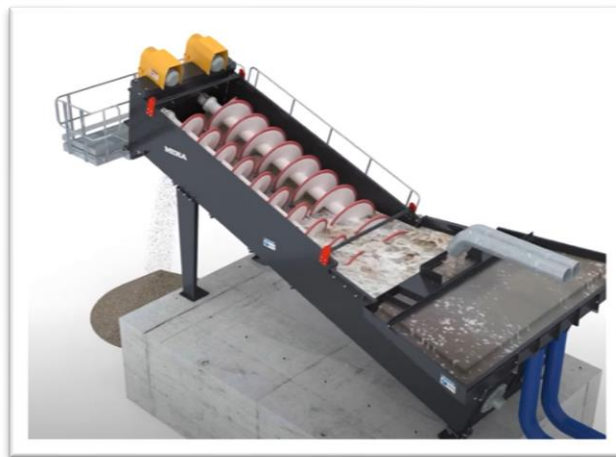
- [1] “www.iqsdirectory.com,” [Online]. Available: <https://www.iqsdirectory.com/articles/baler/compactors.html>. [Accessed 21 07 2024].
- [2] “globaltrashsolutions.com,” [Online]. Available: <https://globaltrashsolutions.com/product-category/compactor/>. [Accessed 21 07 2024].

4 Section 3 – Topic: Gear box for friction washer (210214X)

4.1 Introduction

Friction washer machines are vital in PET recycling, it is used to clear soiled mixed plastics, such as PET flakes by utilizing a high-speed, water-flow mechanism. During a visit to Eco Spindle company for a POM project, we observed that the friction washer machine needed to adjust speed according to the inconsistent supply of plastic bottles.

Most friction washer machines used direct motor control methods to change the speed in the industry. However, this approach results in less effective cleaning, increased machine wear, and higher operating costs. Using a gearbox to control the machine's speed, we can work more efficiently with inconsistent bottle supply rates, ensuring good cleaning performance, reducing mechanical stress, and reducing machine wear



4.2 Acceptance or rejection statement

The project was approved by advisors and also the project was selected as the selected project for the gearbox design project for group 1. The idea was taken from the POM project field visit to ECO-Splinters company as they currently use 4 motors for different production rates. Therefore, by implementing the gearbox configuration for the current plant the initial cost, production efficiency and many other positive impact for the company earnings.

4.3 Design Calculations and Materials Selection

From here onwards, the calculations are done for the project topic that we've selected to continue which is gearbox for fine material washer.

Selected components by 210214X:

- Shaft calculations
- Belt drives and pulleys

4.3.1 Shaft calculations

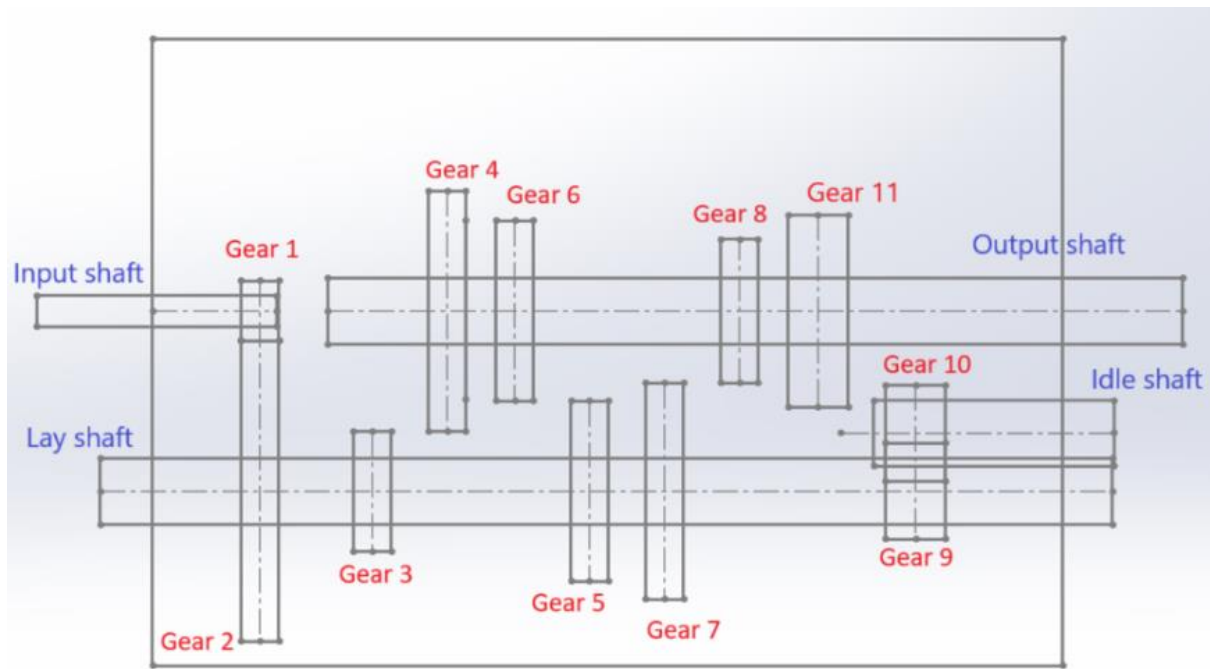
There are 4 shafts in gearbox.

1. Input shaft
2. Lay shaft
3. Output shaft
4. Idle shaft

In this section sequentially calculated torque, tangential load, bending moments, and finally diameter of the shaft.

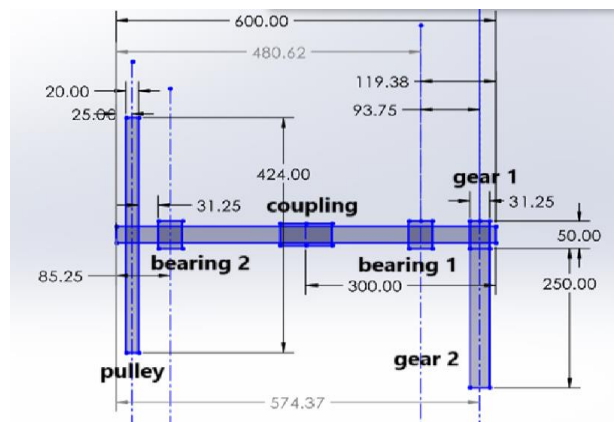
Assumptions:

- we used motor power (2.2 KW) for the calculations to add an additional safe factor. Because this is the maximum power the shaft receives if there are no losses in power transmissions.
- There is no power losses in the gearbox.
- The weight of the shafts is neglected.

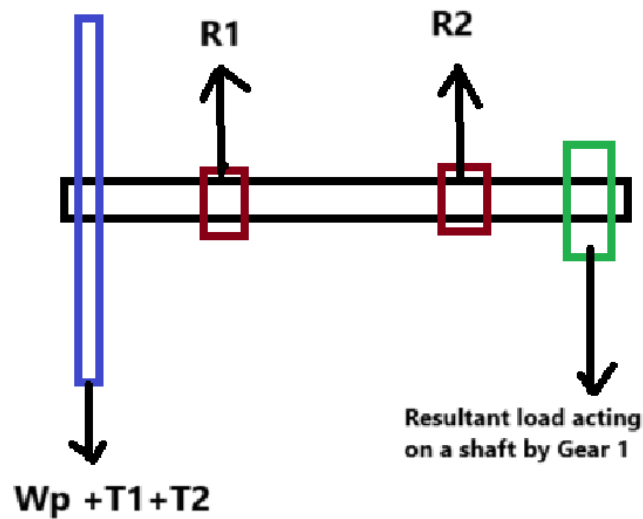


Gears and shafts layout

4.3.1.1 Design the Input Shaft



Distances of input shaft which required to calculate bending moment



Loads acting on input shaft

Calculate the torque of input shaft

$$T = \frac{P}{\omega}$$

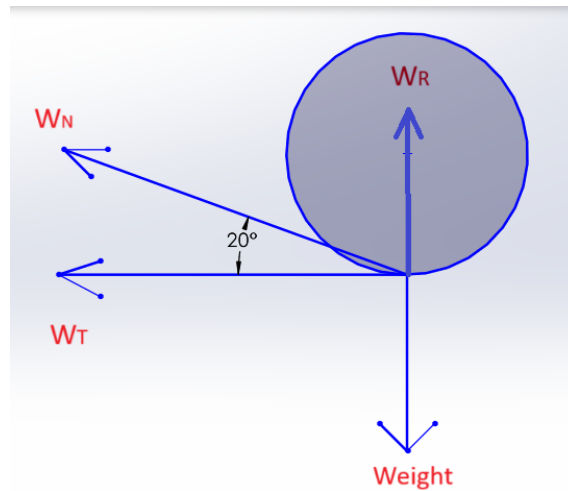
Power transmitted to the gearbox (W)	2200
Angular velocity (rpm)	156.66
Angular velocity (rad/s)	16.40
Torque (Nm)	134.10

Force calculation of input shaft pinion gear

We used 20 full depth involute gears. Therefore, pressure angle 20°.

Pressure Angle (degrees)	20
Pitch-Line Velocity (m/s)	0.410
y	0.108
cv	0.879
Working Stress (MPa)	938.379
Service Factor	1.25
Allowable Tangential Load (N)	24965.922

Tangential Load (without S.F.)(N)	5364.088
Tangential Load (N)	6705.110
Normal Load to the tooth (N)	7135.4296
Density (Avg value for steel)(Kg/m3)	7540
Weight of the Gear (N)	4.538



Horizontal Component of Resultant load = $w_N \times \sin(20) = w_T = 6705.110 \text{ N}$

Vertical Component of Resultant load = $w - w_N \times \cos(20)$

$$= 4.538582821 - 7135.429648 \times \cos(20) = -2435.922 \text{ N}$$

Vertical Component of Resultant load (N)	-2435.922
Horizontal Component of Resultant Load (N)	6705.110

Force calculation of pulley 4

Pulley 4	
material	cast iron
Density (Kg/m3)	7230
Diameter (mm)	424

Width (mm)	20
------------	----

Weight of the pulley 4 = $Density \times \pi \times (radius)^2 \times width$

$$= 7230 \times \pi \times (424)^2 \times 20 \times 10^{-9} = 20.416N$$

Weight of pulley 4	20.416
--------------------	--------

Tension of pulleys obtained using pulley calculation,

Tension in the tight side, T1 (N)	516.79
Tension in the slack side, T2 (N)	69.9

Resultant load by pulley = W + T1 +T2

Vertical Component of Resultant load by pulley (N)	607.107
--	---------

Bending moment calculations

Vertical bending moment



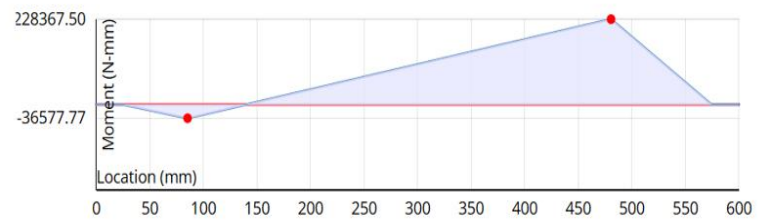
Moment Diagram

(Max +ve)Moment Load (N-mm): 228367.500,

Location (mm): 480.620

(Max -ve)Moment Load (N-mm): -36577.775,

Location (mm): 85.250



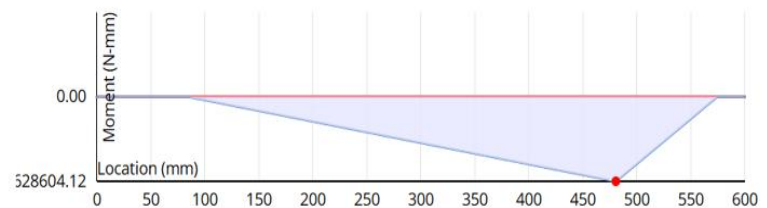
Horizontal bending moment



Moment Diagram

(Max -ve)Moment Load (N-mm): -628604.117,

Location (mm): 480.620



Vertical bending moment (Nmm)	228367.500
Horizontal bending moment (Nmm)	628604.117
Resultant Maximum Bending Moment (Nmm)	668801.054
Resultant Maximum Bending Moment (Nm)	668.801

Recomentded values for Km and Kt

<i>Nature of load</i>	K_m	K_t
1. Stationary shafts		
(a) Gradually applied load	1.0	1.0
(b) Suddenly applied load	1.5 to 2.0	1.5 to 2.0
2. Rotating shafts		
(a) Gradually applied or steady load	1.5	1.0
(b) Suddenly applied load with minor shocks only	1.5 to 2.0	1.5 to 2.0
(c) Suddenly applied load with heavy shocks	2.0 to 3.0	1.5 to 3.0

All shafts rotating and receive a steady load. Therefore,

$$K_m = 1.5$$

$$K_t = 1$$

K_m = combined shock and fatigue factor for bending

K_t = combined shock and fatigue factor for torsion

Calculate the diameters of the shafts, there are two methods.

1. Maximum shear stress theory / guest's theory – used for ductile materials
2. Maximum normal stress theory / Rankine theory – used for brittle materials

Both the theories were applied, and the maximum diameters were chosen.

Equivalent Twisting Moment

$$\begin{aligned}
 \text{Twisting moment } (T_e) &= \sqrt{K_m M^2 + K_t T^2} \\
 &= \sqrt{(1.5 \times 668.8010548)^2 + (1 \times 134.1022117)^2} = 1012.125 \text{ Nm}
 \end{aligned}$$

Equivalent bending moment

$$\text{Equivalent bending moment } (B_e) = \frac{K_m M + \sqrt{K_m M^2 + K_m T^2}}{2}$$

$$= \frac{1.5 \times 668.8010548 + \sqrt{(1.5 \times 668.8010548)^2 + (1 \times 134.1022117)^2}}{2} = 1007.663 \text{ Nm}$$

Equivalent Twisting Moment (Nm)	1012.125
Equivalent Bending Moment (Nm)	1007.663

Material Selection

AISI 4340 Steel (oil quenched 845 C) is selected because it has a high tensile strength and good toughness and impact Resistance, good fatigue strength, high hardness after heat treatment, good machinability and manufacturability and corrosion resistance.

Material	AISI 4340 Steel, oil quenched 845 C
Elastic Limit in Tension	740 MPa
Ultimate Tensile strength	1160 MPa
Permissible Shear Stress (MPa)	Minimum((0.3*740),(0.18*1160))= 208.8 MPa
Permissible Tensile Stress (MPa)	Minimum((0.6*740),(0.36*1160))= 417.6 MPa

We know that equivalent Twisting moment.

$$(T_e) = \frac{\pi}{16} \times \tau \times d^3$$

$\tau = \text{allowable shear}$

$d = \text{diameter of the shaft}$

Permissible Shear Stress (MPa)	Minimum((0.3*1475),(0.18*1595))= 287.1 MPa
Diameter of the Input shaft D (m)	0.026185267
Required Minimum Diameter (mm)	26.185

From above equation we can calculate d = **26.185mm**

we know that equivalent bending moment

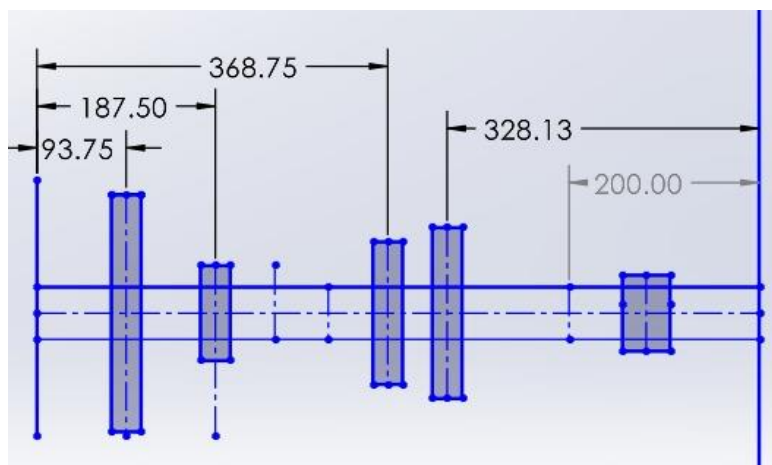
$$B_e = \frac{\pi}{32} \times \sigma_b \times d^3$$

Permissible Tensile Stress (MPa)	Minimum((0.6*1475),(0.36*1595))=574.2 MPa
Diameter of the Input shaft D (m)	0.026146734
Diameter of the Input shaft D (mm)	26.147

From above equation we can calculate d = **26.147mm**

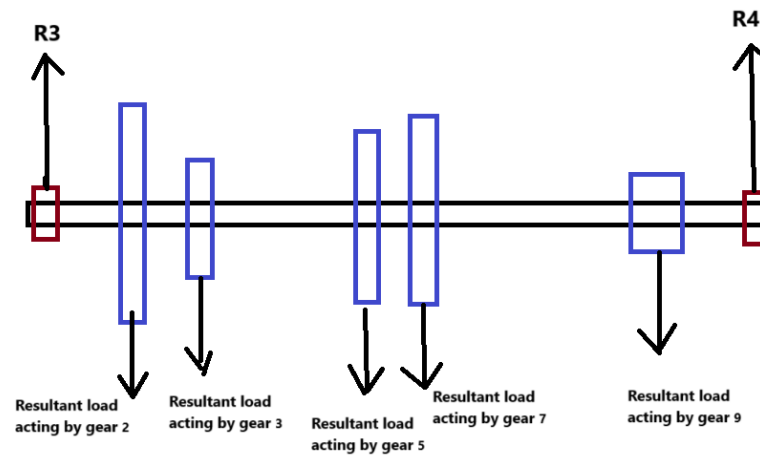
Selected Diameter of the Input shaft (mm)	30
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4.3.1.2 Design the Lay Shaft



Distances of lay shaft which required to calculate bending moment

In lay shaft, when the gears are meshed gear weight and normal load are applied. When gears are not meshed only gear weight applied. Therefore, we need to calculate the bending moment for 1st, 2nd, 3rd, reverse gear and neutral stage.



Loads acting on lay shaft

Calculate the torque of input shaft

Power transmitted to the gearbox (W)	2200
Angular velocity (rpm)	31.33
Angular velocity (rad/s)	3.28.870
Torque (Nm)	670.554

Input gear details and torque calculation

	Clutch Gear	Lay Shaft Gear	1st Gear	2nd Gear	3rd Gear	Reverse Gear
Gear number	1	2	3	5	7	9
RPM	156.66	31.33	31.33	31.33	31.33	31.33
Module (mm)	2.5	2.5	2.5	2.5	2.5	4
Number of teeth	20	100	40	60	72	20
Diameter (mm)	50	250	100	150	180	80
Face Width (mm)	31.25	31.25	31.25	31.25	31.25	50
Material	High Carbon Steel	AISI 4337 Steel	High Carbon Steel	AISI 4337 Steel	AISI 4337 Steel	High Carbon Steel
Maximum UTS of Material	3200	1448	3200	1448	1448	3200
Allowable Static Stress (MPa)	1066.666667	482.6666667	1066.666667	482.6666667	482.6666667	1066.666667
Brinell Hardness	600	435	600	435	435	600
Surface Endurance Limit (MPa)	1610	1148	1610	1148	1148	1610
Flexural Endurance Limit (MPa)	1050	761.25	1050	761.25	761.25	1050
Young's Modulus (GPa)	205	205	205	205	205	205
Power transmitted to the gearbox (W)	2200					
RPM	156.66	31.33	31.33	31.33	31.33	31.33
rad/s	16.40539684	3.280869928	3.280869928	3.280869928	3.280869928	3.280869928
Torque (Nm)	134.1022117	0.380996512	0.380996512	0.380996512	0.380996512	0.380996512

Calculations Tangential loads and Component of vertical and horizontal Resultant load

Pressure Angle (degrees)	20	20	20	20	20	20
Pitch-Line Velocity (m/s)	0.410134921	0.410108741	0.164043496	0.246065245	0.295278294	0.131234797
y	0.1084	0.14488	0.1312	0.1388	0.141333333	0.1084
cv	0.879730588	0.879737342	0.948153843	0.924195841	0.910393518	0.958088484
Working Stress (MPa)	938.3792941	424.6198904	1011.364099	446.0785261	439.4166049	1021.96105
Service Factor	1.25	1.25	1.25	1.25	1.25	1.25
Allowable Tangential Load (N)	24965.92155	15099.01701	32567.26374	15196.39894	15242.66726	69605.48986
Tangential Load (without S.F.)(N)	5364.088469	5364.430894	13411.07723	8940.718156	7450.598464	16763.84654
Tangential Load (N)	6705.110586	6705.538617	16763.84654	11175.8977	9313.24808	20954.80818
Normal Load to the tooth (N)	7135.429648	7135.885149	17839.71287	11893.14192	9910.951596	22299.64109
Density (Avg value for steel)	7540	7540	7540	7540	7540	7540
Weight of the Gear (N)	4.538582821	113.4645705	18.15433128	40.84724539	58.82003336	18.59003524

Resultant Load acting on the Gear (N)	7136.983209	7175.484518	17845.93017	11907.17436	9931.223045	
Angle between resultant force and vertical axis	1.221132903	1.206870749	1.220774545	1.21850688	1.216164894	
Vertical Component of Resultant load	2444.999254	2554.081032	6119.695485	4108.541348	3448.565119	-12071.12148
Horizontal Component of Resultant Load	6705.110586	-6705.538617	16763.84654	11175.8977	9313.24808	18738.00598
Gear number	1	2	3	5	7	9
Position of the gears from LHS	93.75	93.75	187.5	368.75	431.25	640.63

Bending moment Calculation

1st Gear

Gears 1 and 2, and gears 3 and 4 were messed.

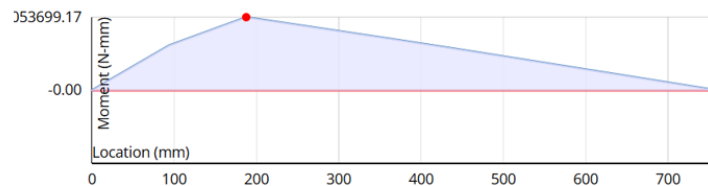
Vertical Bending moment calculations



Moment Diagram

(Max +ve)Moment Load (N-mm): 1053699.168,

Location (mm): 187.500



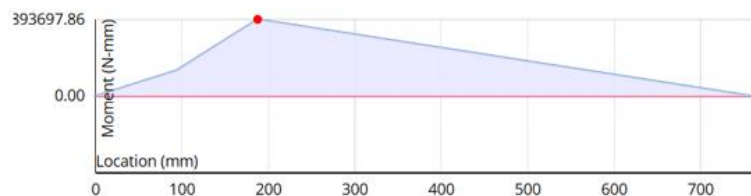
Horizontal Bending moment calculations



Moment Diagram

(Max +ve)Moment Load (N-mm): 1893697.864,

Location (mm): 187.500



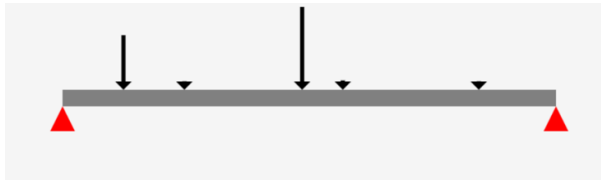
Gear	1ST
Vertical Bending Moment (Nmm)	1053699.168
Horizontal Bending Moment (Nmm)	1893697.864
Resultant Maximum Bending Moment (Nmm)	2167111.796

Resultant Maximum Bending Moment (Nm)	2167.112
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2nd Gear

Gears 1 and 2, and gears 5 and 6 were messed.

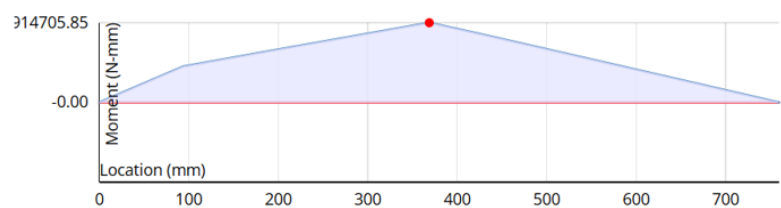
Vertical Bending moment calculations



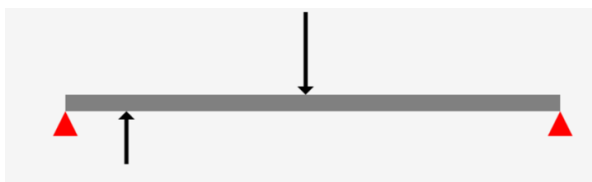
Moment Diagram

(Max +ve)Moment Load (N-mm): 914705.851,

Location (mm): 368.750



Horizontal Bending moment calculations



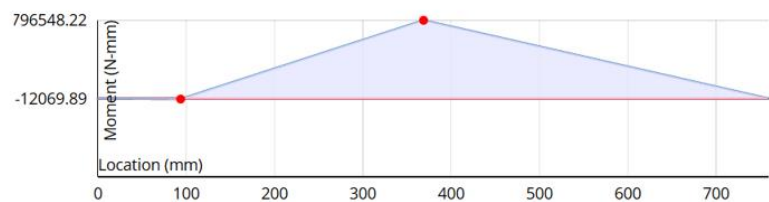
Moment Diagram

(Max +ve)Moment Load (N-mm): 1796548.220,

Location (mm): 368.750

(Max -ve)Moment Load (N-mm): -12069.890,

Location (mm): 93.750



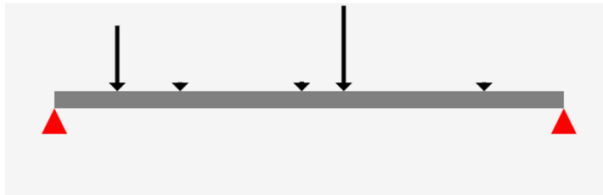
Gear	2ND
Vertical Bending Moment (Nmm)	914705.851
Horizontal Bending Moment (Nmm)	1796548.22

Resultant Maximum Bending Moment (Nmm)	2016004.043
Resultant Maximum Bending Moment (Nm)	2016.004

3rd Gear

Gears 1 and 2, and gears 5 and 6 were messed.

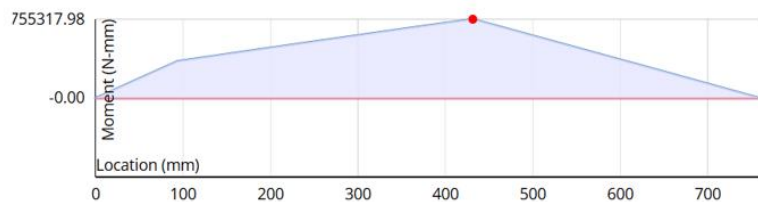
Vertical Bending moment calculations



Moment Diagram

(Max +ve)Moment Load (N-mm): 755317.984,

Location (mm): 431.250



Horizontal Bending moment calculations



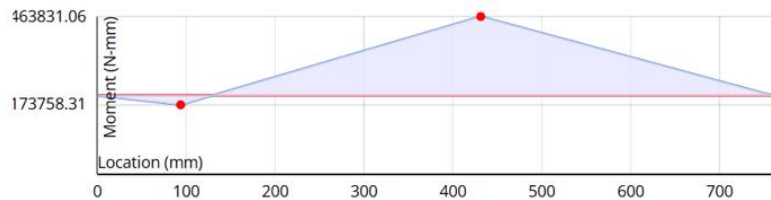
Moment Diagram

(Max +ve)Moment Load (N-mm): 1463831.058,

Location (mm): 431.250

(Max -ve)Moment Load (N-mm): -173758.310,

Location (mm): 93.750



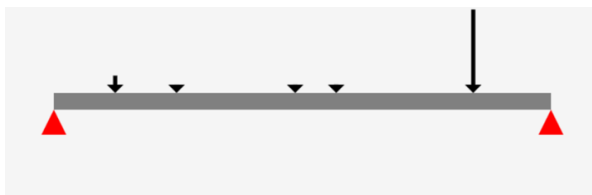
Gear	3RD
Vertical Bending Moment (Nmm)	755317.984
Horizontal Bending Moment (Nmm)	1463831.058
Resultant Maximum Bending Moment (Nmm)	1647211.772

Resultant Maximum Bending Moment (Nm)	1647.212
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Reverse Gear

Gears 1 and 2, and gears 9 and 10 were messed.

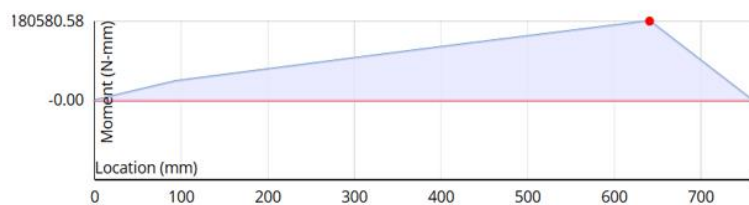
Vertical Bending moment calculations



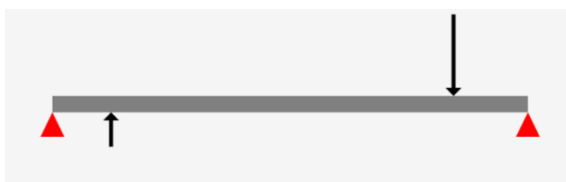
Moment Diagram

(Max +ve) Moment Load (N-mm): 2180580.584,

Location (mm): 640.630



Horizontal Bending moment calculations



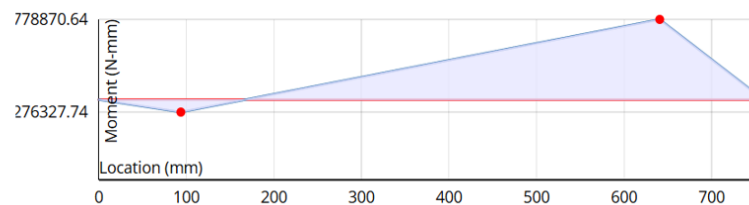
Moment Diagram

(Max +ve) Moment Load (N-mm): 1778870.641,

Location (mm): 640.630

(Max -ve) Moment Load (N-mm): -276327.743,

Location (mm): 93.750



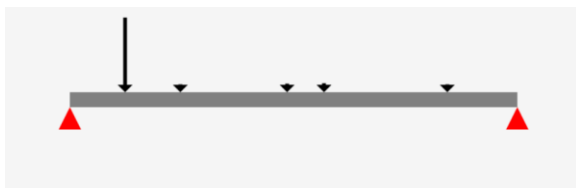
Gear	REV
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Vertical Bending Moment (Nmm)	2180580.584
Horizontal Bending Moment (Nmm)	1778870.641
Resultant Maximum Bending Moment (Nmm)	2814127.296
Resultant Maximum Bending Moment (Nm)	2814.127296

Neutral position

Gears 1 and 2 were messed.

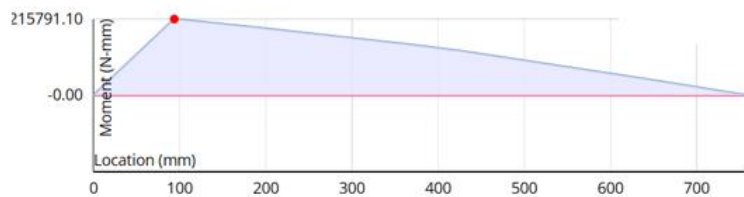
Vertical Bending moment calculations



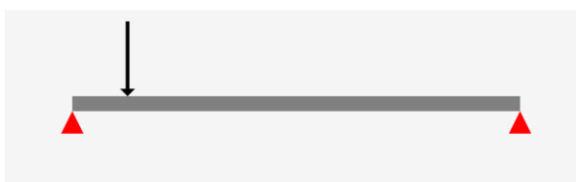
Moment Diagram

(Max +ve)Moment Load (N-mm): 215791.098,

Location (mm): 93.750



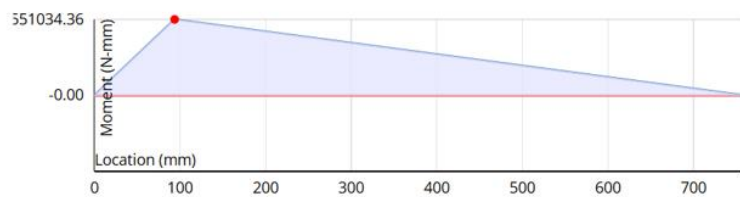
Vertical Bending moment calculations



Moment Diagram

(Max +ve)Moment Load (N-mm): 551034.356,

Location (mm): 93.750



Gear	NEUTRAL
Vertical Bending Moment (Nmm)	215791.098
Horizontal Bending Moment (Nmm)	551034.356
Resultant Maximum Bending Moment (Nmm)	591780.922
Resultant Maximum Bending Moment (Nm)	591.781

Equations to be used in the section below

$$\text{Twisting moment } (T_e) = \sqrt{K_m M^2 + K_m T^2}$$

$$\text{Equivalent bending moment } (B_e) = \frac{K_m M + \sqrt{K_m M^2 + K_m T^2}}{2}$$

Gear	NEUTRAL	1ST	2ND	3RD	REV
Equivalent Twisting Moment (Nm)	1565.098	3969.845	3718.045	3112.319	5062.600
Equivalent Bending Moment (Nm)	1300.357	3881.145	3623.026	2997.469	4993.661

Material Selection

Material	AISI 4340 Steel, oil quenched 845 C
Elastic Limit in Tension	1475 MPa
Ultimate Tensile strength	1595 MPa
Permissible Shear Stress (MPa)	minimum ((0.3*1475),(0.18*1595))= 287.1 MPa
Permissible Tensile Stress (MPa)	minimum ((0.6*1475),(0.36*1595))=574.2 MPa

Diameter calculation

Equivalent Twisting moment.

$$(T_e) = \frac{\pi}{16} \times \tau \times d^3$$

	NEUTRAL	1ST	2ND	3RD	REV
Diameter of the Layshaft D (m)	0.031654402	0.045575683	0.044542067	0.041814941	0.049570075

Diameter of the Layshaft D (mm)	32	46	45	42	50
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Equivalent bending moment

$$B_e = \frac{\pi}{32} \times \sigma_b \times d^3$$

	NEUTRAL	1ST	2ND	3RD	REV
Diameter of the Layshaft D (m)	0.031654402	0.045575683	0.044542067	0.041814941	0.049570075
Diameter of the Layshaft D (mm)	32	46	45	42	50

Selected Diameter of the Lay/Counter shaft (mm)	55
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Output Gear details and torque calculation

	1st Gear	2nd Gear	3rd Gear	Reverse Gear
Gear number	4	6	8	11
RPM	15.665	31.33	46.995	15.665
Module (mm)	2.5	2.5	2.5	4
Number of teeth	80	60	48	40
Diameter (mm)	200	150	120	160
Face Width (mm)	31.25	31.25	31.25	50
Material	High Carbon Steel	AISI 4337 Steel	AISI 4337 Steel	High Carbon Steel
Maximum UTS of Material	3200	1448	1448	3200
Allowable Static Stress (MPa)	1066.666667	482.6666667	482.6666667	1066.666667
Brinell Hardness	600	435	435	600
Surface Endurance Limit (MPa)	1610	1148	1148	1610
Flexural Endurance Limit (MPa)	1050	761.25	761.25	1050
Young's Modulus (GPa)	205	205	205	205
Power transmitted to the gearbox (W)	2200			
rad/s	1.640434964	3.280869928	4.921304892	1.640434964
Torque (Nm)	1341.107723	670.5538617	447.0359078	1341.107723

Calculations Tangential loads and Component of vertical and horizontal Resultant load

Pressure Angle (degrees)	20	20	20	20
Pitch-Line Velocity (m/s)	0.164043496	0.246065245	0.295278294	0.131234797
y	0.1426	0.1388	0.135	0.1312
cv	0.948153843	0.924195841	0.910393518	0.958088484
Working Stress (Mpa)	1011.364099	446.0785261	439.4166049	1021.96105
Service Factor	1.25	1.25	1.25	1.25
Allowable Tangential Load (N)	35397.04123	15196.39894	14559.6232	84245.75895
Tangential Load (without S.F.)(N)	13411.07723	8940.718156	7450.598464	16763.84654
Tangential Load (N)	-16763.84654	-11175.8977	-9313.24808	20954.80818
Normal Load to the tooth (N)	-17839.71287	-11893.14192	-9910.951596	22299.64109
Density (Avg value for steel)	7540	7540	7540	7540
Weight of the Gear (N)	72.61732514	40.84724539	26.14223705	74.36014094
Resultant Load acting on the Gear (N)	17864.67979	11907.17436	9919.923185	22284.18514
Vertical Component of Resultant load (N)	-6028.923828	-4026.846857	-3363.602848	4744.973235
Horizontal Component of Resultant Load (N)	-16763.84654	-11175.8977	-9313.24808	3068.819084
Gear number	4	6	8	11
Position of the gears from LHS	509.38	453.13	265.63	200

Bending moment Calculation

1st Gear

Gears 3 and 4 were messed.

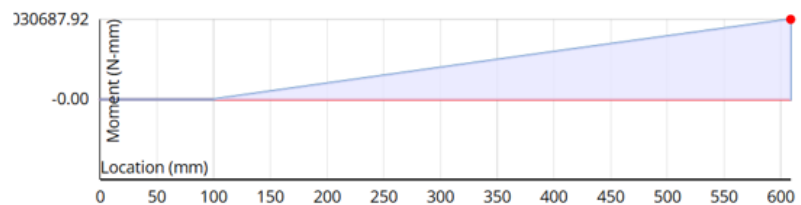
Vertical Bending moment calculations



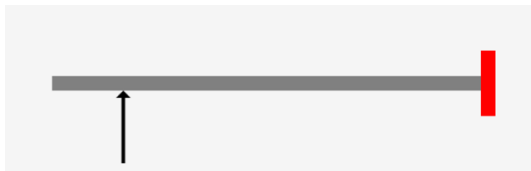
Moment Diagram

(Max +ve)Moment Load (N-mm): 3030687.917,

Location (mm): 608.750



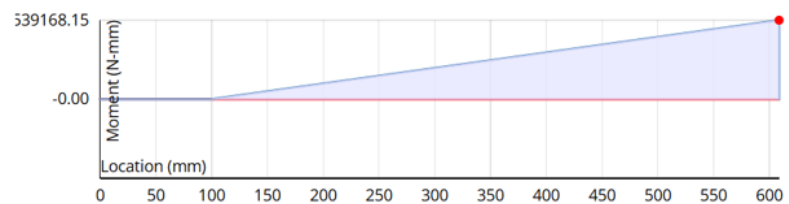
Horizontal Bending moment calculations



Moment Diagram

(Max +ve)Moment Load (N-mm): 8539168.151,

Location (mm): 608.750



Gear	1ST
Vertical Bending Moment (Nmm)	3030687.917
Horizontal Bending Moment (Nmm)	8539168.151
Resultant Maximum Bending Moment (Nmm)	9061040.887
Resultant Maximum Bending Moment (Nm)	9061.041

2nd Gear

Gears 5 and 6 were messed.

Vertical Bending moment calculations



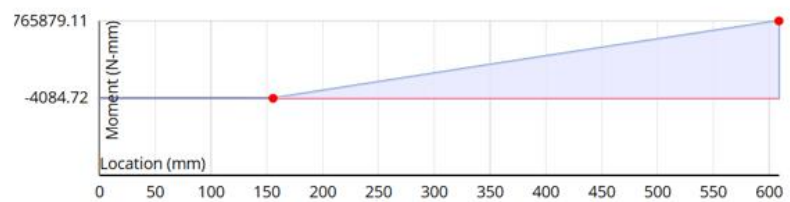
Moment Diagram

(Max +ve)Moment Load (N-mm): 1765879.113,

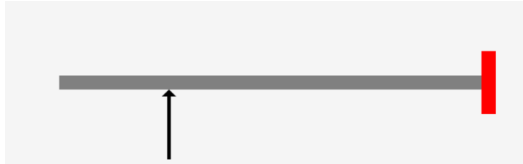
Location (mm): 608.750

(Max -ve)Moment Load (N-mm): -4084.725,

Location (mm): 155.620



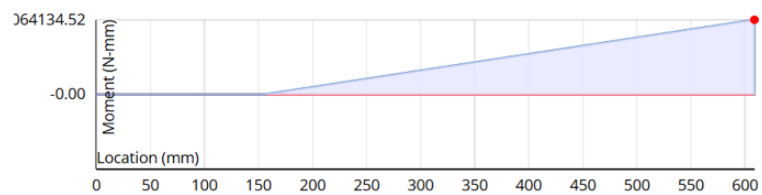
Horizontal Bending moment calculations



Moment Diagram

(Max +ve)Moment Load (N-mm): 5064134.525,

Location (mm): 608.750



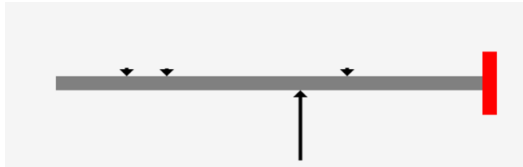
Gear	2ND
Vertical Bending Moment (Nmm)	1765879.113
Horizontal Bending Moment (Nmm)	5064134.525
Resultant Maximum Bending Moment (Nmm)	5363188.187

Resultant Maximum Bending Moment (Nm)	5363.188
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3rd Gear

gears 5 and 6 were messed.

Vertical Bending moment calculations



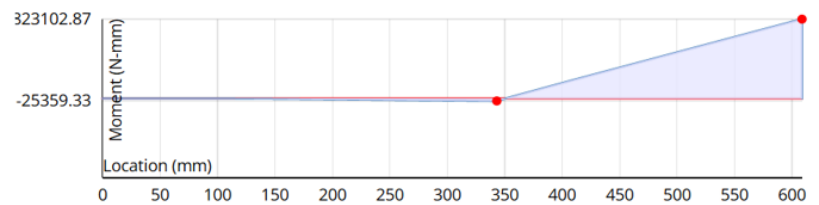
Moment Diagram

(Max +ve)Moment Load (N-mm): 823102.871,

Location (mm): 608.750

(Max -ve)Moment Load (N-mm): -25359.332,

Location (mm): 343.120



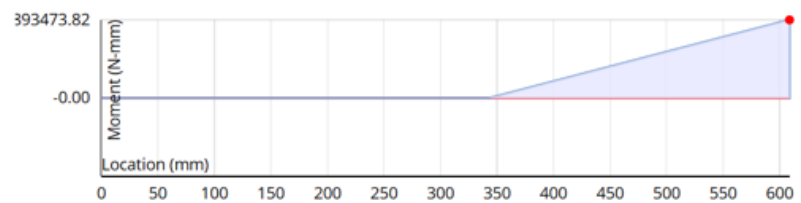
Horizontal Bending moment calculations



Moment Diagram

(Max +ve)Moment Load (N-mm): 893473.825,

Location (mm): 608.750



Gear	3RD
Vertical Bending Moment (Nmm)	823102.871
Horizontal Bending Moment (Nmm)	893473.825
Resultant Maximum Bending Moment (Nmm)	1214822.544

Resultant Maximum Bending Moment (Nm)

1214.823

Reverse Gear

Gears 9 and 10 were messed.

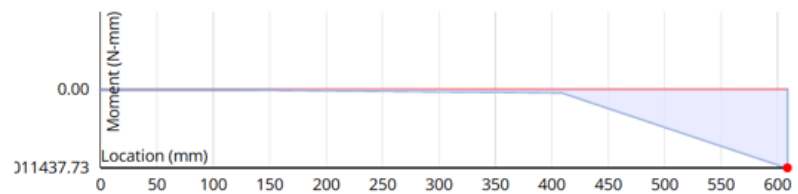
Vertical Bending moment calculations



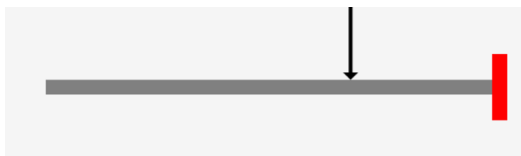
Moment Diagram

(Max -ve)Moment Load (N-mm): -1011437.735,

Location (mm): 608.750



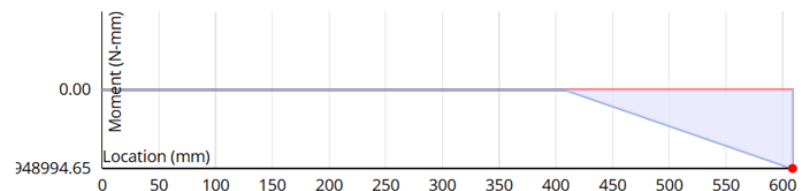
Horizontal Bending moment calculations



Moment Diagram

(Max -ve)Moment Load (N-mm): -948994.647,

Location (mm): 608.750



Gear	REV
Vertical Bending Moment (Nmm)	1011437.735

Horizontal Bending Moment (Nmm)	948994.647
Resultant Maximum Bending Moment (Nmm)	1386938.042
Resultant Maximum Bending Moment (Nm)	1386.938042

Equation to be used in the section below

$$\text{Twisting moment } (T_e) = \sqrt{K_m M^2 + K_m T^2}$$

$$\text{Equivalent bending moment } (B_e) = \frac{K_m M + \sqrt{K_m M^2 + K_m T^2}}{2}$$

Gear	1ST	2ND	3RD	Reverse
Equivalent Twisting Moment (Nm)	13657.566	8072.680	1876.266	2475.209
Equivalent Bending Moment (Nm)	13624.563	8058.731	1849.250	2277.808

Material Selection

Material	AISI 4340 Steel, oil quenched 845 C
Elastic Limit in Tension=	1475 MPa
Ultimate Tensile strength	1595 MPa
Permissible Shear Stress (MPa)	minimum ((0.3*1475),(0.18*1595))= 287.1 MPa
Permissible Tensile Stress (MPa)	minimum ((0.6*1475),(0.36*1595))=574.2 MPa

Diameter calculation

equivalent Twisting moment.

$$(T_e) = \frac{\pi}{16} \times \tau \times d^3$$

Gear	1ST	2ND	3RD	REV
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Diameter of the output shaft D (m)	0.06234047	0.052318051	0.032166996	0.035279004
Diameter of the output shaft D (mm)	62	52	32	35

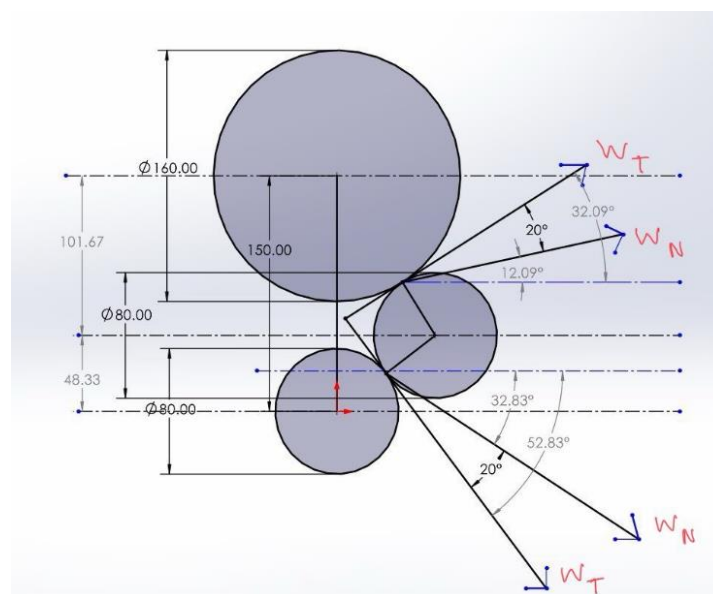
Equivalent bending moment

$$B_e = \frac{\pi}{32} \times \sigma_b \times d^3$$

	1ST	2ND	3RD	REV
Diameter of the Layshaft D (m)	0.062290216	0.0522879	0.032011857	0.034315055
Diameter of the Layshaft D (mm)	62	52	32	34

Selected Diameter of the Output shaft (mm)	65
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4.3.1.3 Design the Idler Shaft



Loads acting on idle gear

Idle Gear details and torque calculation

	Reverse Gear and Idle Gear		
Gear Number	9	10	11
RPM	31.33	31.33	15.665
Module	4	4	4
Number of teeth	20	20	40
Diameter (mm)	80	80	160
Face Width (mm)	50	50	50
Material	High Carbon Steel	High Carbon Steel	AISI 4337 Steel
Maximum UTS of Material	3200	3200	1448
Allowable Static Stress (MPa)	1066.666667	1066.666667	482.6666667
Flexural Endurance Limit (MPa)	1050	1050	761.25
Surface Endurance Limit (MPa)	1610	1610	1148
Brinell Hardness	600	600	435
Young's Modulus (GPa)	205	205	205
Power transmitted (W)	2200		
rad/s	3.280869928	3.280869928	1.640434964
Torque (Nm)	670.5538617	670.5538617	1341.107723

Calculations Tangential loads and Component of vertical and horizontal Resultant load

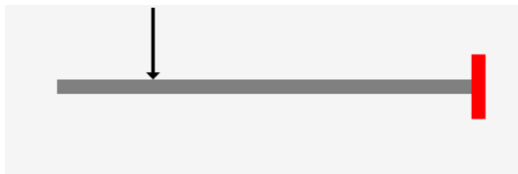
Pressure Angle (degrees)	20	20	20
Pitch-Line Velocity (m/s)	0.131234797	0.131234797	0.131234797
y	0.1084	0.1084	0.1312
Cv	0.958088484	0.958088484	0.958088484
Working Stress (MPa)	1021.96105	1021.96105	462.437375
Service Factor	1.25	1.25	1.25
Allowable Tangential Load (N)	20954.80818	20954.80818	20954.80818
Tangential Load without S.F.(N)	16763.84654	16763.84654	16763.84654
Tangential Load (N)	20954.80818	20954.80818	20954.80818
Normal Load to the tooth(N)	22299.64109	22299.64109	22299.64109
Density of high carbon steel(Kg/m3)	7540	7540	7540
Weight of the Gear (N)	18.59003524	18.59003524	74.36014094
Vertical Component of Resultant load (N)		7437.68846	
Horizontal Component of Resultant Load (N)		-40543.03673	

Bending moment calculation

Reverse Gear

Gears 9, 10, 11 were messed.

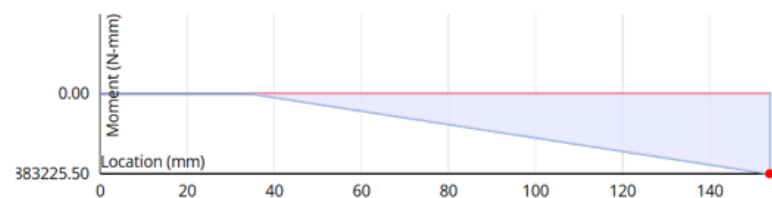
Vertical Bending moment calculations



Moment Diagram

(Max -ve) Moment Load (N-mm): -883225.505,

Location (mm): 153.750



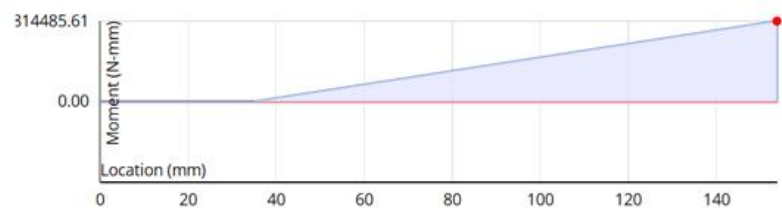
Horizontal Bending moment calculations



Moment Diagram

(Max +ve) Moment Load (N-mm): 4814485.612,

Location (mm): 153.750



Gear	NEUTRAL
Vertical Bending Moment (Nmm)	883225.505
Horizontal Bending Moment (Nmm)	4814485.612
Resultant Maximum Bending Moment (Nmm)	4894829.823
Resultant Maximum Bending Moment (Nm)	4894.830

$$\text{Twisting moment } (T_e) = \sqrt{K_m M^2 + K_m T^2}$$

Equivalent Twisting Moment (Nm)	7372.801
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$$\text{Equivalent bending moment } (B_e) = \frac{K_m M + \sqrt{K_m M^2 + K_m T^2}}{2}$$

Equivalent Bending Moment (Nm)	7357.523
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Material Selection

Material	AISI 4340 Steel, oil quenched 845 C
Elastic Limit in Tension=	1475 MPa
Ultimate Tensile strength	1595 MPa
Permissible Shear Stress (MPa)	minimum((0.3*1475),(0.18*1595))= 287.1 MPa
Permissible Tensile Stress (MPa)	minimum((0.6*1475),(0.36*1595))=574.2 MPa

Diameter calculation

equivalent Twisting moment.

$$(T_e) = \frac{\pi}{16} \times \tau \times d^3$$

Diameter of the Layshaft D (m)	0.05076018
Diameter of the Layshaft D (mm)	50.760

equivalent bending moment

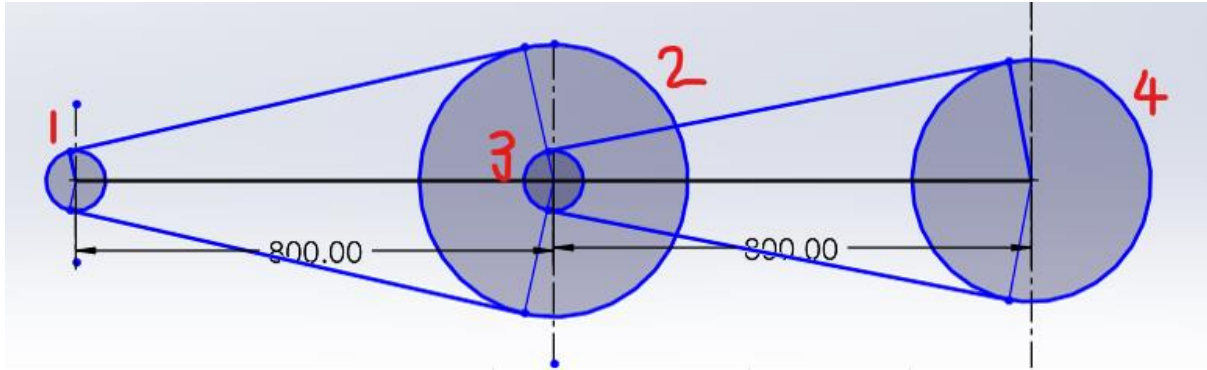
$$B_e = \frac{\pi}{32} \times \sigma_b \times d^3$$

Diameter of the Layshaft D (m)	0.050725093
Diameter of the Layshaft D (mm)	50.725

Selected Diameter of the Lay/Counter shaft (mm)	55
---	----

4.3.2 Belt drives and pulleys

Two pulley and belt drives have been used to obtain the required reduction.



To calculate the shaft, belt tensions must be determined. A V-belt is chosen for its ability to transmit more power with less slipping, due to increased friction.

Motor specs
2.2kW
2820 rpm
7.5 Nm

Pulley Drive 1

Pulley 1 and Pulley 2 are connected via a belt.

Pulley drive 1	
Reduction	4.5:1
Output speed (rpm)	626.67
Output speed (rad/S)	65.62472894

Pulley Drive 2

Pulley 3 and Pulley 5 are connected via a belt.

Pulley drive 2	
Reduction	4.0:1
Output speed	156.66
Output speed (rad/S)	16.40539684

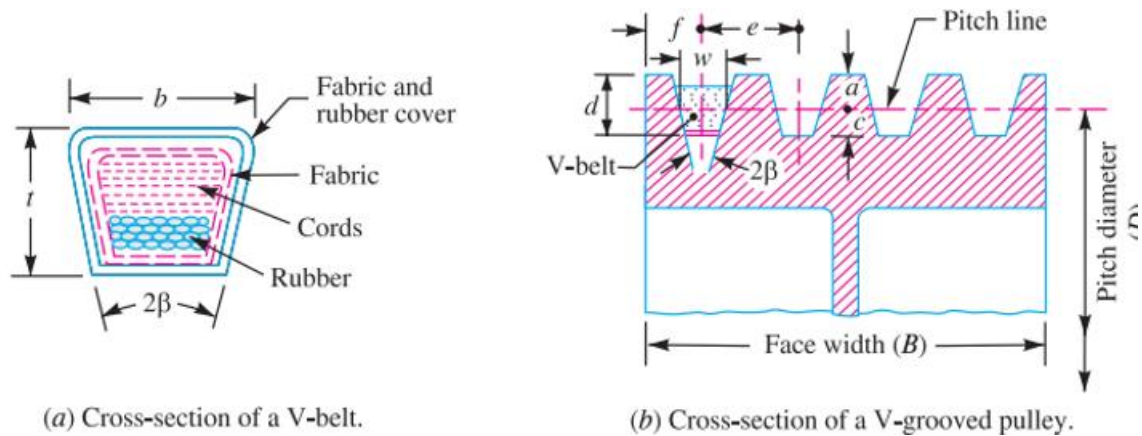


Table 20.1. Dimensions of standard V-belts according to IS: 2494 – 1974.

Type of belt	Power ranges in kW	Minimum pitch diameter of pulley (D) mm	Top width (b) mm	Thickness (t) mm	Weight per metre length in newton
A	0.7 – 3.5	75	13	8	1.06
B	2 – 15	125	17	11	1.89
C	7.5 – 75	200	22	14	3.43
D	20 – 150	355	32	19	5.96
E	30 – 350	500	38	23	–

Table 20.2. Dimensions of standard V-grooved pulleys according to IS : 2494-1974.
(All dimensions in mm)

Type of belt	w	d	a	c	f	e	No. of sheave grooves (n)	Groove angle (2β) in degrees
A	11	12	3.3	8.7	10	15	6	32, 34, 38
B	14	15	4.2	10.8	12.5	19	9	32, 34, 38
C	19	20	5.7	14.3	17	25.5	14	34, 36, 38
D	27	28	8.1	19.9	24	37	14	34, 36, 38
E	32	33	9.6	23.4	29	44.5	20	–

Note : Face width (B) = $(n - 1)e + 2f$

b	13
w	11
d	12
a	3.3
c	8.7
f	10
e	15

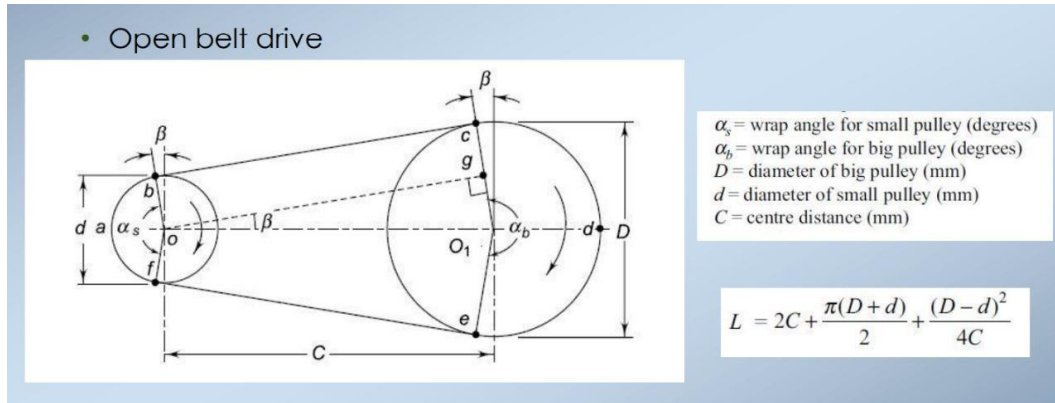
Pulley drive 1	
power range	0.7-3.5
Type of belt	A
Minimum pitch Diameter	75 mm
Diameter of Pulley 1	100 mm

Pulley drive 2	
power range	0.7-3.5
Type of belt	A
Minimum pitch Diameter	75 mm
Diameter of Pulley 3	100 mm

Calculate the Diameter of pulley 2 and 4.

$$N_1 / N_2 = (D_1 + t) / (D_2 + t)$$

Diameter of Pulley 2	478 mm
Diameter of Pulley 4	424 mm



Belt 1	
Length (mm)	2552.571527
Length (m)	2.552571527

Belt 2	
Length (mm)	2455.902275
Length (m)	2.455902275

Weight per meter length = 1.06N

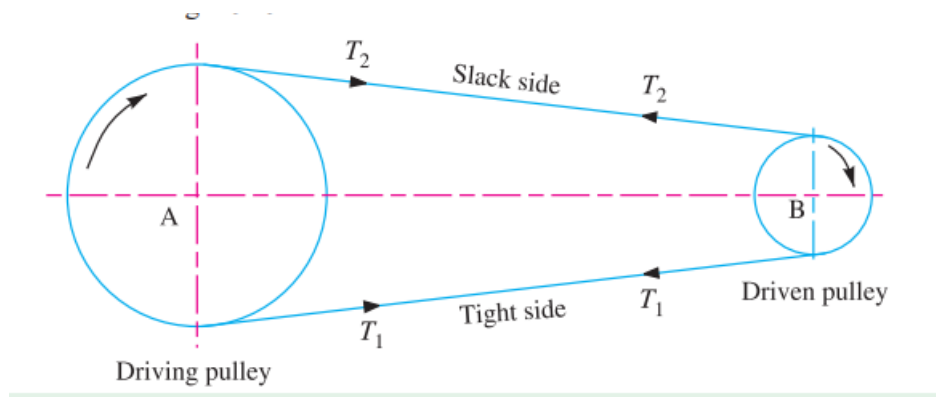
Area of the belt drive, $A = b \cdot t = 104 \text{ mm}^2$

Weight of the belt 1 = 2.705725819 N

Weight of the belt 2 = 2.603256412 N

Mass of the belt 1 = 0.275813029 Kg

Mass of the belt 1 = 0.265367626 Kg



Maximum or total Tension , $T = \sigma A$

Maximum or total Tension in the tight side of belt = 520 N

Centrifugal Tension, $T_c = mv^2$

Centrifugal Tension, of belt 1 $T_c = 67.84933969$ N

Centrifugal Tension, of belt 2 $T_c = 3.209912112$ N

Tension in the tight side belt1, $T_1 = T - T_c = 452.1506603$ N

Tension in the tight side belt2, $T_1 = T - T_c = 516.7900879$ N

Table 18.2. Coefficient of friction between belt and pulley.

Belt material	Pulley material						
	Cast iron, steel			Wood	Compressed paper	Leather face	Rubber face
	Dry	Wet	Greasy				
1. Leather oak tanned	0.25	0.2	0.15	0.3	0.33	0.38	0.40
2. Leather chrome tanned	0.35	0.32	0.22	0.4	0.45	0.48	0.50
3. Convass-stitched	0.20	0.15	0.12	0.23	0.25	0.27	0.30
4. Cotton woven	0.22	0.15	0.12	0.25	0.28	0.27	0.30
5. Rubber	0.30	0.18	—	0.32	0.35	0.40	0.42
6. Balata	0.32	0.20	—	0.35	0.38	0.40	0.42

Coefficient of friction (belt and pulley) = 0.25

$$\alpha = \sin^{-1} \left[\frac{D1 - D2}{2C} \right]$$

Pulley drive 1	
$\sin(\alpha)$	0.23625
α	0.238504783
α	13.66531745
$\theta 1$	152.6693651
$\theta 2$	207.3306349

Pulley drive 2	
$\sin(\alpha)$	0.2025
α	0.20391014
α	11.68319043
$\theta 1$	156.6336191

θ_2	203.3663809
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$$2.3 \log \left[\frac{T_1}{T_2} \right] = \mu \theta \csc \beta$$

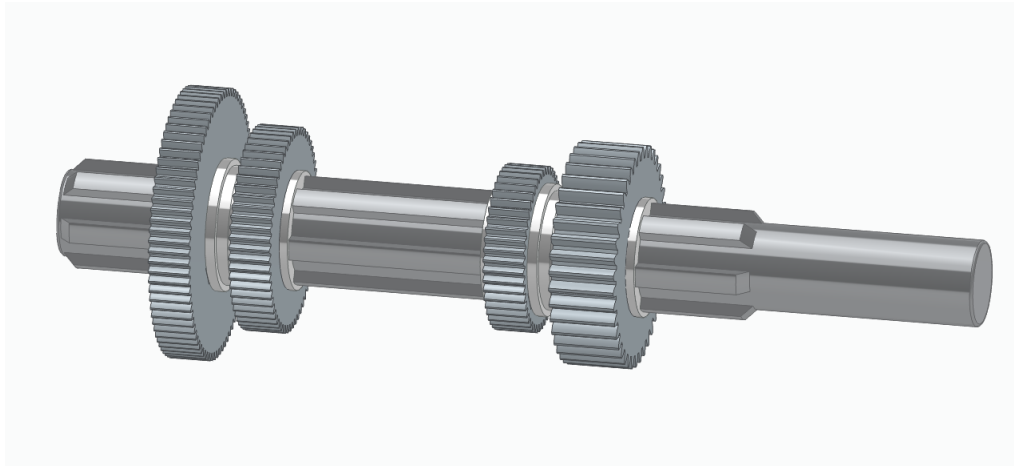
Tension in the Slack side belt1, $T_2 = 64.3416 \text{ N}$

Tension in the Slack side belt2, $T_2 = 69.9086 \text{ N}$

4.4 Final design

Output Shaft

The output shaft is attached to the lay shaft by output gears and lay gears. The output shaft transfer power from the lay shaft to the material washer machine. The single-row roller bearing is used to keep the output shaft inside the gearbox casing. This bearing provides for easy and smooth rotational support, enabling proper alignment of the shaft.



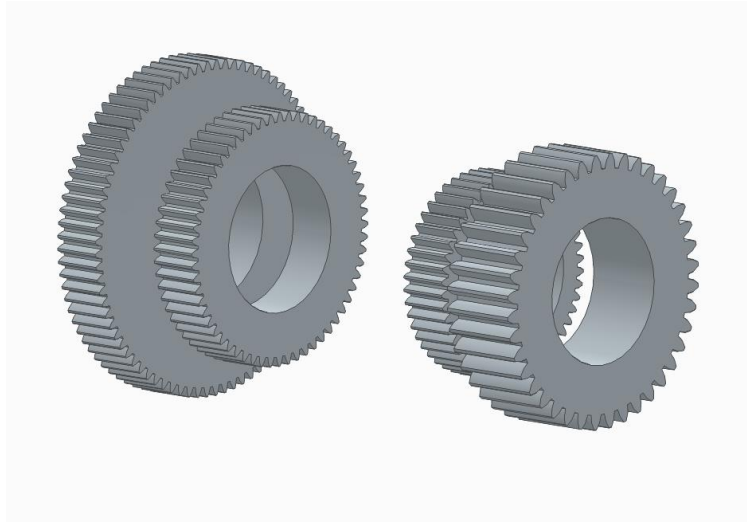
Output Shaft Assembly

The output shaft is machined with splines using milling. There are 6 Splines in output shaft. This spline shaft enables gears to shift smoothly by ensuring that gears can be perfectly engaged and disengaged without slipping.



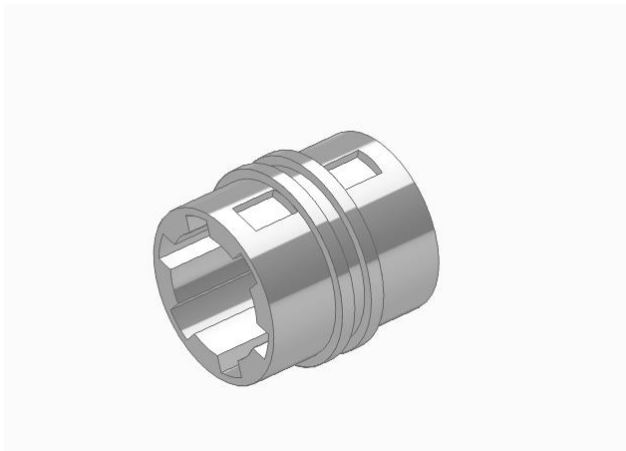
Output Shaft

There 4 gears (gear 4, gear 6, gear 8, gear 11) in output shaft to transfer power.

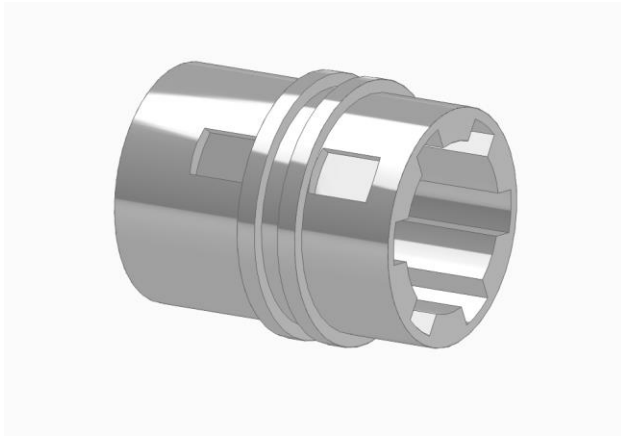


Output Gears

These 4 gears attached to the output shaft by 2 collars. Collars are the important components within the shifting mechanism that help in the connection and locking of gears. Gear 4 and gear 6 attached to the collar 1 and gear 8 and gear 11 attached to the collar 2.

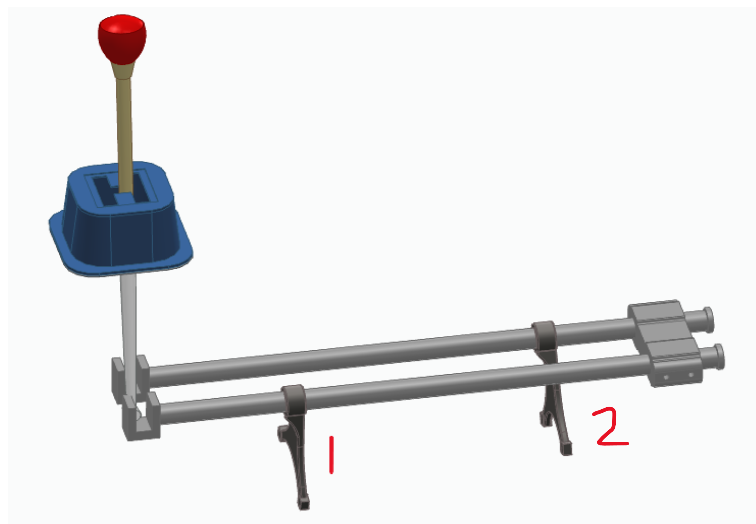


Collar 1



Collar 2

Shifting Mechanism



The shifting mechanism includes two shift forks, two bars, and a shifting lever. Shift Fork 1 connects to Gears 4 and 6, while Shift Fork 2 connects to Gears 8 and 11. Both shift forks are mounted on two bars, which are attached to the gearbox housing by the bars connecting part..

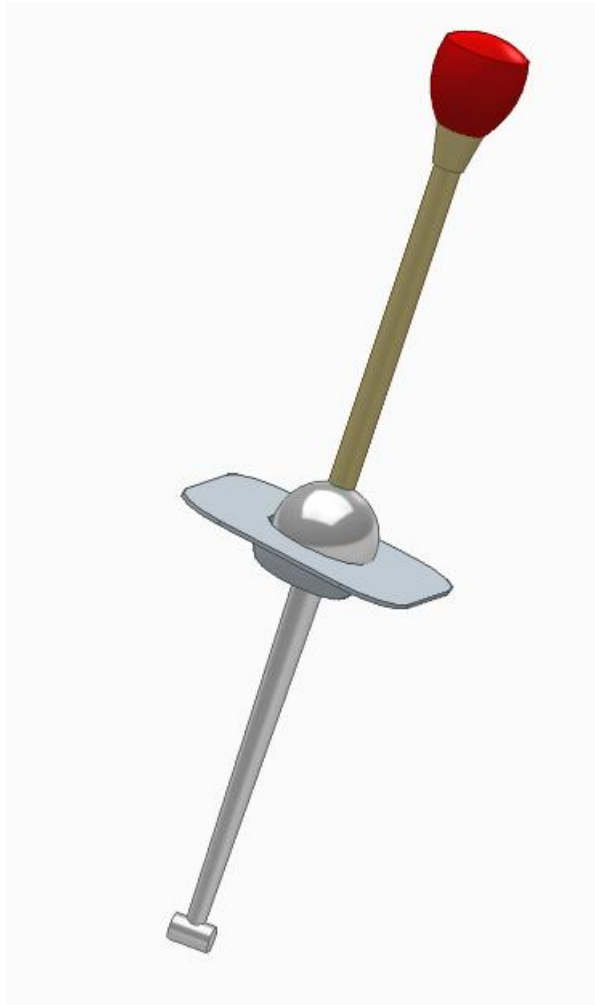


Forks 1 and 2



Fork connecting bars and part that connect to housing

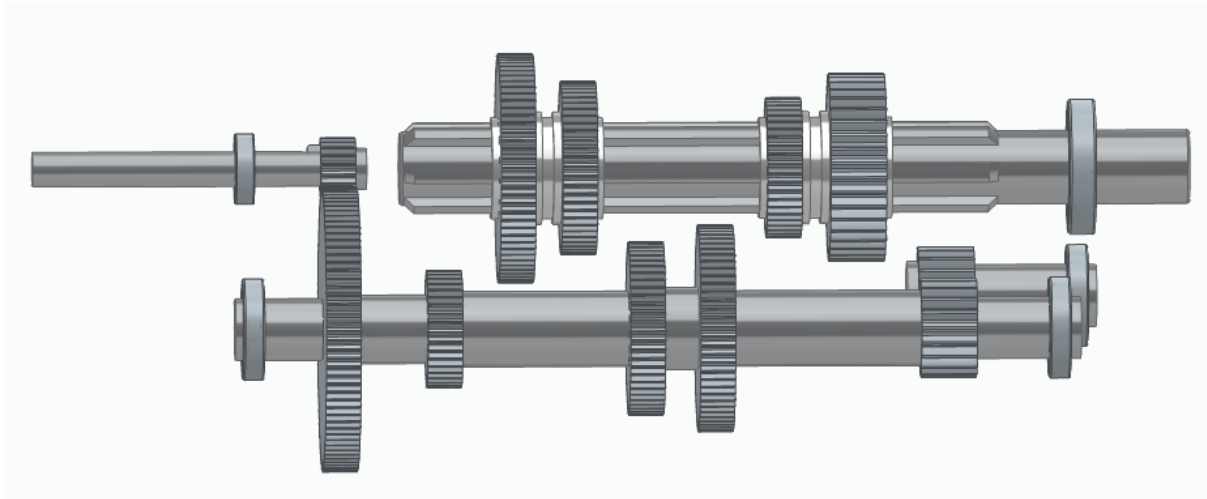
The shifting lever allows the selection of gears in an "H" pattern. The gears are arranged so that the worker can change the gear by moving the lever through a pattern that looks like a letter "H," which makes for easy and quick changing of gears.



Gear lever

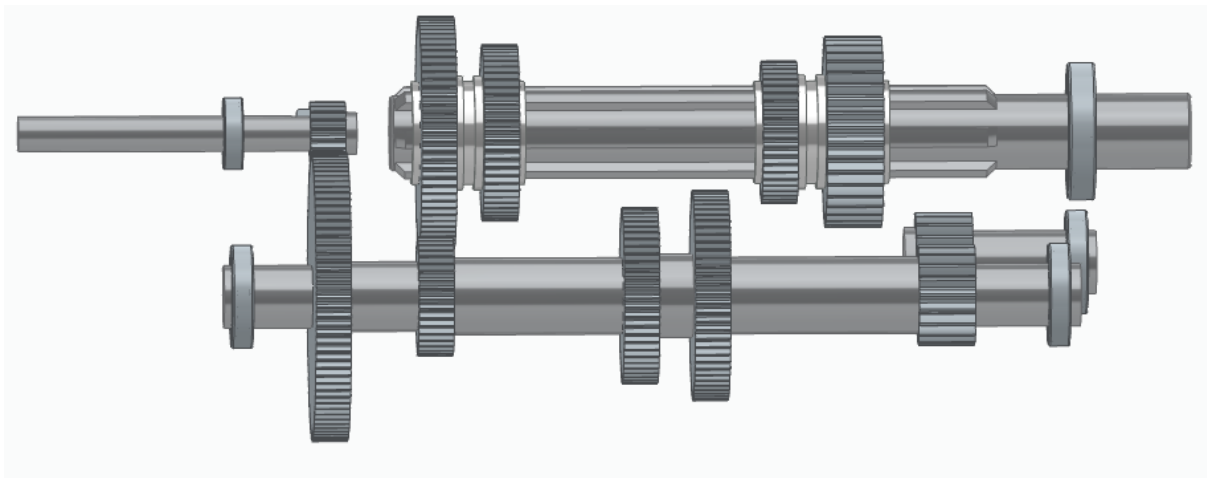
Gear Shifting

Neutral Position



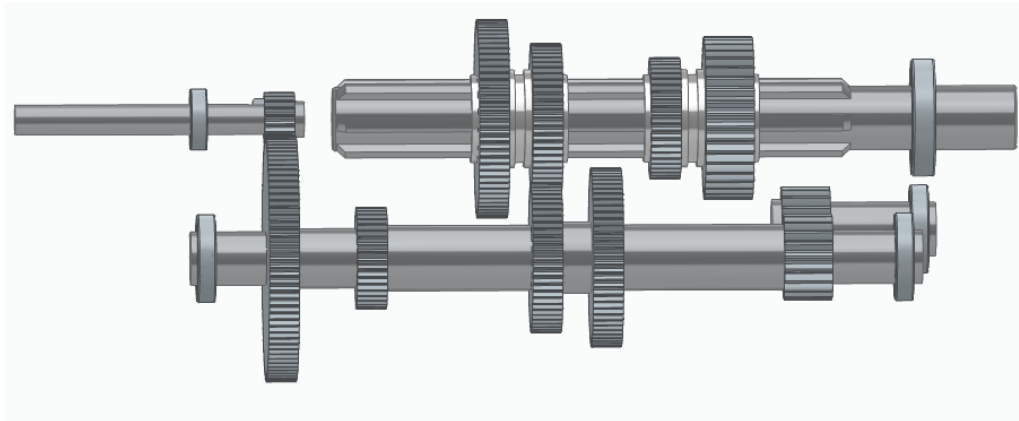
The neutral position input shaft pinion (gear1) has been meshed with input gear in lay shaft (gear 2) and reverse gear (gear 9) in the lay shaft has been meshed with the idler gear (gear 10). These gears are always meshed.

First Gear



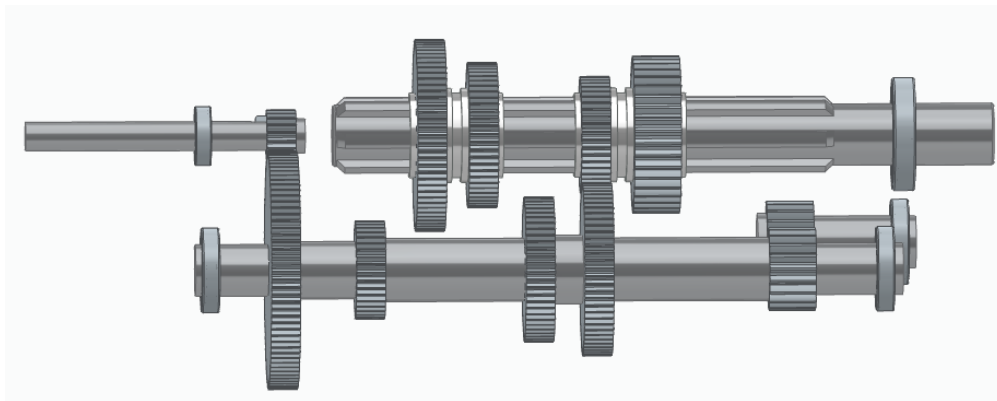
Gear 3 in lay shaft has been meshed with Gear 4 in output shaft.

Second gear



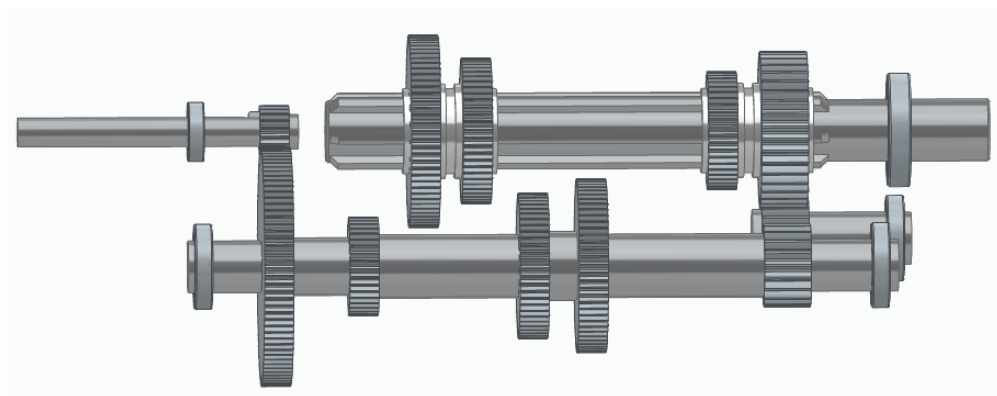
Gear 5 in lay shaft has been meshed with Gear 6 in output shaft.

Third gear



Gear 7 in lay shaft has been meshed with Gear 8 in output shaft.

Reverse Gear



Gear 10 in idle shaft has been meshed with Gear 11 in output shaft.

4.5 Future works

Using topology optimization techniques, we can reduce the weight of the gears for enhance the performance of the gearbox system.

Using topology optimization techniques, we can reduce the weight of the two forks for enhance the performance of the gearbox system.

We can do a detailed calculation for the gear-shifting lever to minimize the effort required for shifting gears. By analysing the force and moment distribution along the lever and evaluating leverage ratios, we can optimize the gear-shifting process.

4.6 References

- [1] “Fine Material Washer Working Principles - MEKA,” [Online]. Available: <https://www.youtube.com/watch?v=jUYk1YCXm4k>. [Accessed 20 7 2024].
- [2] “Friction Washer Applications,” [Online]. Available: <https://www.wiscon-tech.com/friction-washer/>. [Accessed 20 7 2024].
- [3] “High speed friction washer,” [Online]. Available: <https://www.plasticrecyclingmachine.net/high-speed-friction-washer/>. [Accessed 20 07 2024]